SchwingNet: a convolutional-recurrent network for level naming in spine surgery

Article type: Original research

Summary statement: **SchwingNet, trained on rendered CT and radiographs, named lumbar levels on lateral exposures of an unseen C-arm, without a preoperative scan; its transitional naming on real exposures rests on three patients and awaits verified cases.**

Key Points:

- Never trained on the intraoperative C-arm, SchwingNet named the exact vertebra on 595 of 617 detected lateral vertebrae (96.4 %; 95 % confidence interval: 92.3, 98.4) in 141 cases with cone-beam CT truth, against 276 of 631 anteroposterior vertebrae (43.7 %).

- Counting up from the detected sacrum named 979 of 1,009 bodies on anchored lateral and anteroposterior frames (97.0 %); fine-tuned on other cases of that C-arm, SchwingNet named 1,079 of 1,159 lateral vertebrae with three in view (93.1 %) and 608 of 631 anteroposterior (96.4 %), and coupled with the count 991 of 1,009 anchored bodies (98.2 %).

- On the split that chose its checkpoint, its embedding separated the sixth lumbar vertebra from the fifth (silhouette 0.35; 10 persons) more than the fifth from the fourth (0.23); on real exposures its transitional evidence is three cross-validation patients, and certifying 90 % exact-level accuracy needs 35 error-free at-risk persons, and seven real-frame candidates exist.

List of abbreviations: BCa, bias-corrected and accelerated bootstrap interval; C2, second cervical vertebra; CI, confidence interval; GRU, gated recurrent unit; L1–L6, first to sixth lumbar vertebrae; T11–T13, eleventh to thirteenth thoracic vertebrae; t-SNE, t-distributed stochastic neighbour embedding.

Abstract

Purpose: To name vertebral levels from local morphology on one intraoperative C-arm exposure with a network, SchwingNet, test it on an unseen C-arm, and count the transitional persons a test needs.

Materials and Methods: Retrospective study of public collections (analysed April–September 2026; intraoperative exposures November–December 2023): 4,093 CT renders, 9,108 annotated radiographs, 4,276 real exposures with volumetric truth; source median ages 36–66 years, 42.5–64.5 % female, none for the intraoperative, cadaveric or phantom sets. A residual encoder and recurrent head named detected bodies, counted from the detected sacrum and coupled. Naming was scored against cone-beam CT in 148 unseen-C-arm cases. Wilson bounds set transitional sample size against pre-stated targets. Intervals were cluster-adjusted; P values, Holm-adjusted exact tests.

Results: In the embedding (checkpoint-selection split), the sixth lumbar vertebra was farther from the fifth (silhouette 0.35; 95 % confidence interval: 0.31, 0.39) than the fifth from the fourth (0.23; 0.21, 0.25). Never trained on the C-arm, SchwingNet named 595 of 617 detected lateral vertebrae (96.4 %; 92.3, 98.4) and 276 of 631 anteroposterior (43.7 %; $P < .001$); fine-tuned by case-grouped folds, 601 of 617 (97.4 %) and 608 of 631 (96.4 %). On anchored frames the sacrum count named 979 of 1,009 bodies (97.0 %), the coupled call 991 (98.2 %). No real-exposure transitional test person was verified; certifying 90 % needs 35 error-free at-risk persons; seven real-frame candidates exist.

Conclusion: SchwingNet named ordinary lumbar levels from one lateral exposure of an unseen C-arm; its transitional naming on real exposures rests on three cross-validation patients and awaits verified cases.

Introduction

A vertebral level is not marked; it is counted: fluoroscopy shows three or four bodies, and surgeons count up from the sacrum or down from the second cervical vertebra (C2) across exposures [1]. Half of responding spine surgeons report a wrong-level operation [2], a registry of 34,544 consecutive patients recorded 0.12 % (thoracic 0.42 %, lumbar 0.13 %) [3], and the level first exposed is wrong in 4.3–15 % of lumbar operations before a check film corrects it [4,5]. No neurophysiological check confirms a thoracic level: the standard motor examination tests no key muscle between the first thoracic and second lumbar segments [6], and thoracic screw stimulation, recorded from the rectus abdominis, cannot separate segments [7]. A transitional vertebra — a sixth lumbar (L6) or thirteenth thoracic (T13) vertebra, or a fifth lumbar vertebra (L5) fused to the sacrum (Figure 1) — changes the count: 8.8 % of 3,147 full-length radiographs lacked 24 presacral segments [8], and in 550 lumbar operations the count was wrong in 38 of 71 patients with a lumbosacral transitional vertebra [9].

Intraoperative level identification rests on registration: LevelCheck projects a reader's preoperative CT labels onto the radiograph [10,11]. Its published accuracies are scored against those labels: 100 % label placement [12], and 97.6 %, five surgeons' median on 62 difficult films with the overlay [13]. On a transitional spine the imported label is only as right as the planning read, and lumbar MRI localisers read without a count from C2 identified the segments in 77.9–82.2 % [14]. Registration-free naming on real C-arm images imported identity from a preoperative radiograph (73.3 %) [15] or needed long-length (Long-Film) images of cadavers (88.3 %) [16].

We hypothesised that each transitional vertebra carries local morphology that a network reading a few bodies can learn, so a level can be named from one C-arm exposure without counting from C2 or importing a label. The purpose was to build that network (SchwingNet), examine what it learns, test it on an unseen C-arm, and count the transitional persons a test of it needs and the public sources hold.

Materials and Methods

Design. This retrospective study used de-identified image collections (Table 1), analysed from April 2026, when the first collection was downloaded, to 24 September 2026; intraoperative exposures carry study dates 11 November–12 December 2023. The institutional human research protection office determined this secondary analysis not to be human participant research (determination 2026-269); consent

and the Health Insurance Portability and Accountability Act do not apply. Every subject was reported by its release or article; CTSpinoPelvic1K also in a data descriptor [17].

Sources and transitional phenotypes. Rendered CT came from CTSpinoPelvic1K [18] (801 of 802 records), TotalSegmentator v2 and v3 [19] and VerSe 19/20 [20]; CSXA [21], BUU-LSPINE [22] and NHANES II [23] supplied radiographs with released levels or landmarks, and VinDr-SpineXR pseudo-labelled films [24]. Real exposures with three-dimensional truth came from C-ArmLSpine [25] (296 exposures, 148 of 149 cases, one mobile C-arm, per-case cone-beam CT), DeepFluoro (six specimens [26]), COBRA2026 (145 of 467 Elekta cases; seven of 152 downloaded failed a pre-export check [27]), a phantom [28] and a cadaveric specimen [29] (Table 1; flow, Figure S1). Transitional persons were typed under a rib convention as L6, T13, sacralised L5 (Castellvi III or IV [30]), Castellvi I/II or lumbar rib (Figure 1, Table 2): VerSe from its published read; CTSpinoPelvic1K from two radiology residents (post-graduate year 2) grading independently, six disagreements resolved by consensus [17]; TotalSegmentator by an independent rib count; intraoperative and COBRA2026 calls are an automatic labeller's (SPINEPS with VERIDAH [31]), read by no person.

Reference standard. No person read an intraoperative image. Truth is the per-case cone-beam CT, registered to each exposure and labelled by TotalSegmentator and by SPINEPS with VERIDAH; a vertebra is scored only where both agree (814 of 910), withholding T11 and most of T12, and both count from the sacrum, so agreement is not independent. Naming matches detections one to one to vertebral-body truth boxes (Table 3); detection is scored against whole-vertebra projections.

The level encoder and what it learns. Each body is read from a fixed-scale two-channel crop (image and target marker; target 96 of 320 pixels) by a ResNet-18 encoder [32] into a 512-dimensional embedding. The latest encoder's embedding was mapped by t-distributed stochastic neighbour embedding (t-SNE) on its validation split after dropping 2,889 crops of 164 persons also in its training keys (Figure 2); that split chose the checkpoint, so it is not held out. Separation from the level above is the cosine silhouette, bootstrapped by person and repeated per render source. The three contested C-ArmLSpine L6 patients, in neither split, and the last free vertebra of validation persons with an unlabelled fused L5 were placed against the neighbouring clusters (supplement). Occlusion over ten lateral crops per level from persons outside the training keys gave the share and density of attribution on the body above, the target and the body below. Renders were styled by image-formation terms fitted per device, which missed their

pre-stated criterion (Figures S2 and S3, Table S1).

The network. Two detectors never trained on the test C-arm find the bodies (Figure 3): a YOLO11m-pose keypoint detector trained on films boxes the bodies the level model names (confidence 0.35), and a YOLO11m vertebra and sacrum detector, fine-tuned on real frames of two other devices, boxes the sacrum at confidence 0.74, chosen on three other devices (Table S2); 97 % detection was pre-stated as counting grade. A two-layer bidirectional gated recurrent unit (GRU) decodes runs of two to nine embeddings at mixed lengths into a 27-level posterior. Counting ranks each body from the detected sacrum and reads no morphology. The coupled call chooses, among the ordinary, L6, T13 and both-variant naming alphabets, the one best fitting the posteriors at the ordinals; a classifier name differing from the call, or an alphabet margin below the fifth percentile of correct frames in the other folds, stops the read (Table 3).

Naming, counting and coupling on the intraoperative C-arm. C-ArmLSpine was scored zero-shot by the latest encoder and its recurrent head, whose training lists hold no C-ArmLSpine case; with five case-grouped folds; and, because a case is an acquisition session rather than a proven person (five dates), with three folds by acquisition day. Per fold that encoder and head were fine-tuned (15 epochs, learning rate 10^{-4}), checkpoints chosen on validation folds excluding every test case (Figure 4). The primary outcome, designated after a first zero-shot run was scored, was the exact level of detected lateral vertebrae at full coverage; secondary outcomes were naming with one or three bodies in view, the count, the coupled call and its stop signal, and detection on the 56 test frames (Table S2, Figure S4). Same-device detection was measured on nested whole-case subsets of 0 to 210 real frames (two seeds; supplement). Detection against view angle used $\pm 90^{\circ}$ render pose pools per device and one patient's 301-frame rotational acquisition, the real pool holding only anteroposterior and lateral views (Figure S5). Naming against training persons was measured on NHANES II, BUU-LSPINE and CSXA with a render-only encoder on nested person subsets. Render-only detector arms at matched count compared anteroposterior-lateral pose pools, an arc of poses and the pools doubled (two devices, two seeds); a nested inverse-power fit of accuracy against count, with and without a lateral-angle entropy term (supplement), was compared by AICc and likelihood ratio.

Transitional sample size. Pre-stated Wilson targets, a 95 % lower bound of 90 % for the exact level and 85 % for variant sensitivity on pooled at-risk test persons, set the persons a transitional test needs (Table 2, Figure 5); at risk are the subtypes a count from the sacrum or from C2 misnames: L6, T13

and a sacralised L5 on a four-lumbar count. The person-disjoint split (50:10:40 per subtype) puts every real transitional subject on the test side; of its transitional test persons only the three C-ArmLSpine L6 patients, scored in cross-validation, enter a result here.

Statistical analysis. Naming used every available case. Transitional sample size is the smallest person count at which the Wilson 95 % lower bound with zero to five errors reaches a target (Figure 5a). Intervals are 95 % two-thousand-resample bootstraps clustered by case, patient or person, percentile with the bias-corrected and accelerated (BCa) interval beside; where errors fall in few clusters, the Wilson interval on the design-effect-adjusted count replaces it. Paired comparisons report the same-item difference with its bootstrap interval and an exact sign test over clusters, Holm-adjusted over four comparisons named before adjustment (lateral versus anteroposterior naming; three in the supplement); other contrasts are exploratory; significance was two-sided $P < .05$. Every reported number passed a patient-keyed evaluation gate that counts a model's, and its ancestors', training and checkpoint-selection persons as seen (supplement, with the other integrity checks). Analyses used Python 3.11 and 3.12 (Python Software Foundation, Wilmington, Del), PyTorch 2.6.0 (PyTorch Foundation), Ultralytics 8.4 (Ultralytics, Frederick, Md), NumPy 2.2.6 and SciPy 1.15.3 and 1.18.1; reporting follows CLAIM 2024 [33].

Results

Study population. The corpus held 4,093 renders of labelled CT, 9,108 level-annotated films and 4,276 real exposures with three-dimensional truth, 778 from mobile C-arms (Table 1). Of the intraoperative set's 910 cone-beam CT vertebrae, 814 (89.5 %) were agreed, giving 676 agreed lateral vertebrae in 143 cases and 673 anteroposterior in 143. No age, sex or indication is released for the intraoperative, cadaveric or phantom sets. Across the eight releases that give them (Table S3), median age ran from 36 years (CSXA) to 66 (COBRA2026) and the share female from 42.5 % (TotalSegmentator) to 64.5 % (CSXA).

Sources and transitional phenotypes. The sources held 194 transitional persons (Table 2): 110 with L6 (three intraoperative), seven with T13, 28 with a sacralised L5 (12 with four lumbar vertebrae), 31 with Castellvi I/II, 14 with a lumbar rib and four unassigned. No released radiograph or C-arm frame carries a human-read T13 or Castellvi grade (Figure 1).

The level encoder and what it learns. On the checkpoint-selection split (13,009 crops of 1,612 persons), the nearest crop from another person was the same level for 9,740 (74.9 %; 95 % confidence interval [CI]: 72.2, 77.6) (Figure 2a, b). L6 separated from L5 (silhouette 0.35; 0.31, 0.39; 98 crops, 10 persons; leave-person-out 0.34) more than L2–L5 from the level above (0.16–0.23; L5 from L4, 0.23; 0.21, 0.25) and as far as L1 from T12 (0.32; 0.29, 0.35); the classifier named 75 of 98 L6 crops and 111 of 1,012 L5 crops L6 (Figure 2c; per render source, supplement). In three never-seen patients the contested L6 sat nearer the L6 cluster under whole-vertebra boxes only, and in two of four fused-L5 validation persons the last free vertebra was named L5, the name a count gives (supplement). No T13 crop was outside training. Per unit area, occlusion attribution was densest on the target body at each of six levels (enrichment 1.39–4.51), but for L4–L6 more fell on the body below (36–43 %) than on the target (24–32 %) (Figure 2d).

Naming, counting and coupling on the intraoperative C-arm. The keypoint detector boxed 617 of 676 agreed lateral bodies (91.3 %), the held-out detector a sacrum box on 233 of the 286 frames showing one (81.5 %) (Figure 4a). Never trained on C-ArmLSpine, SchwingNet named 595 of 617 detected lateral vertebrae exactly (96.4 %; 92.3, 98.4) and 276 of 631 detected anteroposterior (43.7 %), a paired difference of 54.3 points (45.6, 62.9; $P < .001$) on 593 vertebrae of 140 cases (Table 3). Fine-tuned by case-grouped folds, it named 601 of 617 lateral (97.4 %; 93.2, 99.0) and 608 of 631 anteroposterior vertebrae (96.4 %; 92.5, 98.3), 1.0 point (-1.9, 4.0) and 52.6 points (44.1, 61.3) over zero-shot; by acquisition-day folds, 568 of 617 lateral (92.1 %; 87.5, 96.0), 4.4 points below zero-shot (-9.1, 0.2). With three lateral bodies in view the case-grouped models named 1,079 of 1,159 (93.1 %; 89.2, 96.2), with one 451 of 603 (74.8 %) (Figure 4b); zero-shot, 970 of 1,159 (83.7 %) and 385 of 603 (63.8 %). On the 1,009 bodies of anchored frames (confidence 0.74), the count from the sacrum was exact on 979 (97.0 %; 94.2, 99.2) (Figure 4c), the fine-tuned classifier on 986 (97.7 %) and the coupled call on 991 (98.2 %; 96.5, 99.5), 1.2 points over the count (0.0, 3.2) and 0.5 over the classifier (-1.5, 2.4). The read stopped on 23 of 233 anchored frames, catching five of six erring frames and all four L6 calls: three on five-lumbar frames, one on the four anchored frames of the three patients one labeller reads six-lumbar (Figure 4d). The held-out detector found 231 of 287 vertebrae (80.5 %; 74.1, 86.8) and 43 of 56 sacra (76.8 %), short of the pre-stated 97 % (Table S2). Against vertebral-body boxes at confidence 0.74 (supplement), the same-device series' zero-frame detector (styled renders on its geometry plus other C-arms' frames) found 7.0 % of test bodies (20 of 287); 50 frames of 25 cases gave 82.9 % (238), and 50 to 210

frames 81.5–86.8 % (234–249), intervals overlapping, renders posed on all 148 cases' geometry.

Transitional sample size. With no error, 35 at-risk persons put the Wilson lower bound at the pre-stated 90 % and 22 carriers the variant-sensitivity bound at 85 % (Figure 5a). No real-frame transitional test person was verified: the three C-ArmLSpine L6 patients are scored only in cross-validation (Table 3), and against 91 unread COBRA2026 test-side screen calls (Table 2), a two-labeller count with both junctions in the field found seven at-risk persons with an exported frame, whose error-free Wilson lower bound would be 64.6 % (Figure 5a). The split's 51 at-risk CT test persons (36 rendered; most in the current encoder's training data, Table 2) would give 93.0 % error-free.

Persons, films and coverage. With a render-only encoder, exact-level naming on films rose with training persons (Figure 5b): NHANES II 77.8, 87.8 and 92.7 % at five, 25 and all 264 persons; BUU-LSPINE 65.0, 76.7 and 96.9 % (2,967); CSXA 94.2, 96.4 and 99.8 % (4,114). Only CSXA came within two points of its fitted asymptote, from 200 persons; NHANES II and BUU-LSPINE sat -3.2 and -3.1 points from theirs at all persons. Transfer between corpora peaked at 74.3 % (NHANES II to CSXA, 749 films). At matched count (2,646 renders of 125 subjects), Surgio-B51 render-only detectors trained on an arc of poses found 23.6 and 15.1 points more of the unseen-sweep bodies (13,932 boxes) than anteroposterior-lateral pools (two seeds), doubling the pools 13.3 and 7.3 (Figure 5c); the gain was on obliques (39.1, 26.0 points), and recall on the device's real spot frames fell (4.9 to 0.7 %; 9.1 to 1.0 %). The Elekta XVI arms scored zero. Over 105 points, count plus coverage beat count alone (AICc lower by 26.0; likelihood ratio 29.1, $P < .001$; +20.0 points per unit lateral-angle entropy, 12.2 to 27.7; Figure 5d) with the arms read at confidence 0.25, chosen after 0.74 left them few detections; at 0.74 the coverage term added nothing ($P = .82$).

Discussion

Levels are counted across partial exposures, and a transitional vertebra makes the count and an imported label disagree; SchwingNet names a level from one C-arm exposure by the shapes of the bodies in view. Never trained on the intraoperative C-arm, it named 96.4 % of detected lateral vertebrae exactly, and counting from the detected sacrum was exact on 97.0 % of anchored bodies; the anteroposterior view did not survive the unseen device until fine-tuned on it (96.4 %); the lateral did not measurably change.

The encoder set a sixth lumbar vertebra apart from the fifth as far as the rib boundary sets L1 apart from T12, and per unit area drew most on the target body, as local identity predicts. The evidence is thin: ten persons on a selection split, an order reversed in one render source, a contested L6 in three never-seen patients holding under one box convention only, and in two of four fused-L5 persons the name a count gives.

On real exposures the transitional evidence here rests on three C-ArmLSpine patients whose sixth lumbar vertebra one labeller reads and the other does not, scored in cross-validation and agreeing with that read on one of five bodies; that is a description, not a test of transitional naming. The powered comparison with registration carrying a planning read, on verified at-risk persons against a comparator fixed before scoring, is a separate study. No accuracy here is a wrong-level surgery rate.

Detection, not naming, lost most to the unseen device, and same-device frames restored it only to a plateau reached by 50 frames. Every real frame of that device is anteroposterior or lateral, so frames after the fiftieth added no view (three of 18 angle bins, supplement). Render-only arms of that device varying pose alone at matched count found more bodies on held-out oblique poses and on one real rotational acquisition, and fewer on the device's real spot frames; the pooled fit preferred count plus coverage on one device, and only at a confidence below the operating point reported here. Coverage is thus a held-out-angle effect shown on renders, not an account of the plateau on real frames. A level model trained from ImageNet weights on the split's training persons saturates likewise in development: doubling those persons from 4,477 to 8,950 moved validation accuracy on real films by 0.1 points and on rendered L6 bodies by 1.7 (supplement). Styled renders stayed separable from real frames (supplement); paired domain adaptation with 140 paired images cut a synthetic-trained pose regressor's real-radiograph failure rate from 16.1 % to 11.2 % [34], a route untried here.

This study has limitations. One intraoperative device is held, and its truth is two automatic labellers agreeing, which removes the thoracolumbar junction and cannot settle a transitional count; whether its 148 cases are 148 people cannot be established, so every intraoperative interval, treating cases as independent, is too narrow if they are not. Day-grouped fine-tuning scored below zero-shot laterally. The embedding is of a split that chose the checkpoint; most transitional test persons of the built split were in the current encoder's lineage, so no model reported here scores them. Held-out detector numbers use whole-vertebra boxes; the posterior is uncalibrated; nothing has been evaluated prospectively.

In summary, SchwingNet named ordinary lumbar levels on lateral exposures of an unseen C-arm, and on its selection split its embedding set a sixth lumbar vertebra apart from the fifth. Its transitional naming on real exposures is unverified: three cross-validation patients carry the only real transitional exposures scored, certifying 90 % needs 35 error-free at-risk persons, and seven candidates exist. The clinical role is decision support abstaining visibly beside the surgeon's count, never replacing the check film.

Other information

Data availability. Every image source is listed in Table 1 with its licence and download source (flow of cases in Figure S1 of the supplement); VinDr-SpineXR requires a PhysioNet data use agreement, and the literature panels are not redistributed. The derived frame layout is not redistributed because the licences differ; it is rebuilt from the sources by the released exporters. The fitted image-formation parameters are released as one file per device. Result files behind every number, with the path of each number inside them, are listed in the repository's claims ledger and in the file NUMBERS.md.

Code and model availability. Code, the fitted device parameters and the trained weights are in a private repository (branch L6_Class) available to the reviewers and editors on request; a public deposit with a revision identifier and a registered weight hash will accompany publication. The weights are released for retrospective analysis and for bench, phantom and cadaver evaluation; nothing here has been evaluated prospectively.

Ethics. Secondary analysis of de-identified imaging, determined by the institutional human research protection office not to be human participant research (determination 2026-269). C-ArmLSpine and COBRA2026 are used under their own releases (CC BY 4.0 and CC BY-NC 4.0), the Geneva phantom under CC BY-SA 4.0, and the DeepFluoro and Ljubljana cadaveric sets under CC BY-NC-SA 4.0 and CC BY-NC-ND. VinDr-SpineXR was obtained under its PhysioNet data use agreement and is covered by a separate determination of the same office (**[PENDING G0: VinDr determination number and date]**); it trained the intraoperative fold models only (Table 1). The literature fluoroscopy panels are figures of published articles, used for training only and not redistributed.

Registration. Not a clinical trial; not registered.

Funding and acknowledgments. Stated on the full title page, which is submitted separately for double-

anonymized review.

Use of writing tools. Drafting, figure and table assembly and the number-tracing checks were assisted by a large language model operating on the result files; responsibility for the content rests with the authorship stated on the full title page.

References

1. Strong MJ, Santarosa J, Sullivan TP, et al. Pre- and intraoperative thoracic spine localization techniques: a systematic review. J Neurosurg Spine 2022;36(5):792–799. doi: 10.3171/2021.8.SPINE21480.
2. Mody MG, Nourbakhsh A, Stahl DL, Gibbs M, Alfawareh M, Garges KJ. The prevalence of wrong level surgery among spine surgeons. Spine (Phila Pa 1976) 2008;33(2):194–198. doi: 10.1097/BRS.0b013e31816043d1.
3. Furuya M, Fujimori T, Takenaka S, et al. Wrong-level spinal surgery: a multicentre analysis of 34,544 consecutive cases. Bone Joint J 2026;108-B(5):673–680. doi: 10.1302/0301-620X.108B5.BJJ-2025-1129.R2.
4. Dablouk MO, Sajjad J, Lim C, Kaar G, O'Sullivan MGJ. Intra-operative imaging for spinal level localisation in lumbar surgery. Br J Neurosurg 2019;33(3):352–356. doi: 10.1080/02688697.2018.1562030.
5. Ammerman JM, Ammerman MD, Dambrosia J, Ammerman BJ. A prospective evaluation of the role for intraoperative x-ray in lumbar discectomy. Predictors of incorrect level exposure. Surg Neurol 2006;66(5):470–473. doi: 10.1016/j.surneu.2006.05.069.
6. Kirshblum SC, Burns SP, Biering-Sorensen F, et al. International standards for neurological classification of spinal cord injury (revised 2011). J Spinal Cord Med 2011;34(6):535–546. doi: 10.1179/204577211X13207446293695.
7. Kulik G, Pralong E, McManus J, Debatisse D, Schizas C. A CT-based study investigating the relationship between pedicle screw placement and stimulation threshold of compound muscle action potentials measured by intraoperative neurophysiological monitoring. Eur Spine J 2013;22(9):2062–2068. doi: 10.1007/s00586-013-2754-0.

8. Sabri SA, Laynes RA, Hoffman CJ, et al. Implications of lumbosacral transitional anatomy on vertebral numbering: evaluation of 3147 adult full-length spine radiographs. Neurosurg Focus 2025;58(6):E13. doi: 10.3171/2025.3.FOCUS2555.

9. Nakagawa T, Hashimoto K, Tsubakino T, Hoshikawa T, Inawashiro T, Tanaka Y. Lumbosacral transitional vertebrae cause spinal level misconception in surgeries for degenerative lumbar spine disorders. Tohoku J Exp Med 2017;242(3):223–228. doi: 10.1620/tjem.242.223.

10. Otake Y, Schafer S, Stayman JW, et al. Automatic localization of vertebral levels in x-ray fluoroscopy using 3D–2D registration: a tool to reduce wrong-site surgery. Phys Med Biol 2012;57(17):5485–5508. doi: 10.1088/0031-9155/57/17/5485.

11. Lo SFL, Otake Y, Puvanesarajah V, et al. Automatic localization of target vertebrae in spine surgery: clinical evaluation of the LevelCheck registration algorithm. Spine (Phila Pa 1976) 2015;40(8):E476–E483. doi: 10.1097/BRS.0000000000000814.

12. De Silva T, Lo SFL, Aygun N, et al. Utility of the LevelCheck algorithm for decision support in vertebral localization. Spine (Phila Pa 1976) 2016;41(20):E1249–E1256. doi: 10.1097/BRS.0000000000001589.

13. Manbachi A, De Silva T, Uneri A, et al. Clinical Translation of the LevelCheck Decision Support Algorithm for Target Localization in Spine Surgery. Ann Biomed Eng 2018;46(10):1548–1557. doi: 10.1007/s10439-018-2099-2.

14. Peh WC, Siu TH, Chan JH. Determining the lumbar vertebral segments on magnetic resonance imaging. Spine (Phila Pa 1976) 1999;24(17):1852–1855. doi: 10.1097/00007632-199909010-00017.

15. Zhang R, Hu Y, Zhang K, et al. VDVM: an automatic vertebrae detection and vertebral segment matching framework for C-arm X-ray image identification. J Xray Sci Technol 2023;31(5):935–949. doi: 10.3233/XST-230025.

16. Huang Y, Jones CK, Zhang X, et al. Multi-perspective region-based CNNs for vertebrae labeling in intraoperative long-length images. Comput Methods Programs Biomed 2022;227:107222. doi: 10.1016/j.cmpb.2022.107222.

17. Schwing G, Schehr A, Tekumulla A, et al. CTSpinoPelvic1K: spine, pelvis, ribs and femora in one coordinate frame, annotated for lumbosacral transitional anatomy. arXiv:2609.22760 [cs.AI]. 2026. doi: 10.48550/arXiv.2609.22760. Posted September 19, 2026.

18. Schwing G, Schehr A, Tekumulla A, et al. CTSpinoPelvic1K: spine, pelvis, ribs and femora in one coordinate frame, annotated for lumbosacral transitional anatomy. Zenodo, version v11. 2026. doi: 10.5281/zenodo.22797014. Published September 18, 2026.

19. Wasserthal J, Breit HC, Meyer MT, et al. TotalSegmentator: robust segmentation of 104 anatomic structures in CT images. Radiol Artif Intell 2023;5(5):e230024. doi: 10.1148/ryai.230024.

20. Sekuboyina A, Husseini ME, Bayat A, et al. VerSe: a vertebrae labelling and segmentation benchmark for multi-detector CT images. Med Image Anal 2021;73:102166. doi: 10.1016/j.media.2021.102166.

21. Ran Y, Qin W, Qin C, et al. A high-quality dataset featuring classified and annotated cervical spine X-ray atlas. Sci Data 2024;11(1):625. doi: 10.1038/s41597-024-03383-0.

22. Klinwichit P, Yookwan W, Limchareon S, Chinnasarn K, Jang JS, Onuean A. BUU-LSPINE: a Thai open lumbar spine dataset for spondylolisthesis detection. Appl Sci (Basel) 2023;13(15):8646. doi: 10.3390/app13158646.

23. Mo Y, Papież BW, Chen Y, Mandal I, Jabbar F. NHANES II spinal X-ray with extra labels for reference vertebral bodies (C2, S1). Zenodo. 2023. doi: 10.5281/zenodo.10223910. Published November 30, 2023.

24. Nguyen HT, Pham HH, Nguyen NT, et al. VinDr-SpineXR: a deep learning framework for spinal lesions detection and classification from radiographs. *Medical Image Computing and Computer Assisted Intervention – MICCAI 2021*. Ed. by de Bruijne M, Cattin PC, Cotin S, et al. Vol. 12905. Lecture Notes in Computer Science. Cham: Springer, 2021, pp. 291–301. doi: 10.1007/978-3-030-87240-3_28.

25. Feng Q. C-ArmLSpine. Zenodo. 2026. doi: 10.5281/zenodo.22768651. Published September 17, 2026.

26. Grupp RB, Unberath M, Gao C, et al. Automatic annotation of hip anatomy in fluoroscopy for robust and efficient 2D/3D registration. Int J Comput Assist Radiol Surg 2020;15(5):759–769. doi: 10.1007/s11548-020-02162-7.

27. Thummerer A, Rit S, Kamp F, et al. COBRA2026: a large-scale multicenter pelvic CBCT projection dataset. Zenodo, version 1.0.0. 2026. doi: 10.5281/zenodo.21322350. Published July 13, 2026.

28. Schmid J, D'Isidoro F, Chênes C, Ferguson S. 2D-3D registration gold-standard dataset for the hip joint based on uncertainty modeling. Université de Genève, Yareta. 2021. doi: 10.26037/yareta:2kw5s2jadbbv3ngtkzsovihtnu.

29. Tomaževič D, Likar B, Pernuš F. "Gold standard" 2D/3D registration of X-ray to CT and MR images. *Medical Image Computing and Computer-Assisted Intervention – MICCAI 2002*. Ed. by Dohi T, Kikinis R. Vol. 2489. Lecture Notes in Computer Science. Berlin, Heidelberg: Springer, 2002, pp. 461–468. doi: 10.1007/3-540-45787-9_58.

30. Castellvi AE, Goldstein LA, Chan DP. Lumbosacral transitional vertebrae and their relationship with lumbar extradural defects. Spine (Phila Pa 1976) 1984;9(5):493–495. doi: 10.1097/00007632-198407000-00014.

31. Möller H, Schoen H, Graf R, et al. VERIDAH: solving enumeration anomaly aware vertebra labeling across imaging sequences. arXiv:2601.14066 [cs.CV]. 2026. doi: 10.48550/arXiv.2601.14066. Posted January 20, 2026.

32. He K, Zhang X, Ren S, Sun J. Deep residual learning for image recognition. *2016 IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*. 2016, pp. 770–778. doi: 10.1109/CVPR.2016.90.

33. Tejani AS, Klontzas ME, Gatti AA, et al. Checklist for Artificial Intelligence in Medical Imaging (CLAIM): 2024 Update. Radiol Artif Intell 2024;6(4):e240300. doi: 10.1148/ryai.240300.

34. Zheng J, Miao S, Wang ZJ, Liao R. Pairwise domain adaptation module for CNN-based 2-D/3-D registration. J Med Imaging (Bellingham) 2018;5(2):021204. doi: 10.1117/1.JMI.5.2.021204.

Figure Legend(s)

1 Transitional phenotypes. (a) Sixth lumbar, 54-year-old man; (b) thirteenth thoracic, 24-year-old woman; (c) sacralised L5, own label, Castellvi IV, 89-year-old, sex other; (d) four lumbar, L5 in sacrum, Castellvi IIIb, 60-year-old man. Rows: labelled CT (sagittal; b, coronal slab; c, surface); render; another patient's film (women, 31, 64). Arrow and red, transitional vertebra; bar, 20 mm; hatched, none human-read. 19

2 Level-encoder embedding, validation crops (checkpoint-selection split). (a) t-SNE (t-distributed stochastic neighbour embedding) by level; triangles, sixth lumbar vertebra (L6); diamonds, last free vertebra above fused L5; stars, one labeller's L6 in never-seen patients; rings, their other vertebrae; T13, thirteenth thoracic. (b) Lumbosacral window. (c) Silhouette; CI, confidence interval. (d) Occlusion enrichment: attribution over area share; ten lateral crops per level. 20

3 SchwingNet; shaded blocks learned, sizes in pixels. (a) Two detectors, neither naming a level; one boxes bodies, one the sacrum. (b) Level model: two-channel crop per body, ResNet-18, bidirectional gated recurrent unit (GRU) across bodies, 27-level posterior; red, T13 (thirteenth thoracic) and L6 (sixth lumbar). (c) Ordinals counted from the detected sacrum. (d) Posteriors choose the alphabet; outlined, the call. 21

4 Reading C-arm exposures. (a) Bodies (blue) and sacrum (dashed) boxed by detectors never trained on this C-arm. (b) Field b, enlarged, named by shape, fine-tuned on other cases ("none": no name). (c) Ordinals from the sacrum, ordinary alphabet. (d) A frame one CT labeller reads six-lumbar: shape fits the L6 alphabet (outlined); a stop, not a verdict. L1–L6, lumbar vertebrae. 22

5 Persons, films, coverage. (a) Wilson 95 % lower bound by errors against transitional test persons; lines, targets. (b) Naming accuracy against training persons per film corpus; bars, seed range; dashed, tested on NHANES II (NHANES II: on BUU-LSPINE). (c) Render-only detectors at matched N, low- against high-coverage poses. (d) Observed against fitted accuracy: N only (open), N plus coverage (filled). 23

Table 1: Image sources: one row per collection, with its modality and device, its role, the unit the split keeps disjoint, what was scored, its truth and what this work releases from it.

Collection[j]	Modality; device	Role in this work	Distinct subjects (split unit)	Frames, films or renders (train / val / test)	Truth source	Released from it[k]
CT with labels, rendered						
CTSpinoPelvic1K	abdominopelvic CT	level bank[a]	801 patients of 802 records[b] (TCIA patient identifier)	801 renders	release labels: levels, ribs, pelvis	corpus; level weights
TotalSegmentator v2 and v3	whole-body CT	level bank; styled corpus (142 subjects, 17 validation)	1,935 scans[c] (CT scan)	1,170 (v2) and 1,735 (v3) renders	labels mapped by name	corpus; level weights
VerSe 19/20	spine CT	level bank; no accuracy quoted[c]	355 subjects, 374 series (subject)	387 renders; 374 soft-tissue renders	VerSe masks C1–L6; published transitional table	corpus; level weights
Radiographs						
CSXA	cervical lateral	level bank; checkpoint selection	4,963 individuals (individual)	4,963 films	per-vertebra levels, released	corpus; level weights
BUU-LSPINE	lumbar AP and lateral	level bank; checkpoint selection	3,600 cases[d] (case number)	3,603 lateral films; the 3,600 AP films in no bank	end-plate lines L1–S1; levels by position	corpus; level weights
NHANES II	lateral survey film	level bank; checkpoint selection	364 persons among the annotated (sample person)	543 annotated films (542 on disk) of 17,075 release entries[e]	landmarks, released	corpus; level weights
VinDr-SpineXR	AP and lateral	intraoperative fold banks only, training, pseudo-labelled[f]	5,000 studies[c] (study)	2,071 films as 2,632 crops, of 10,466 released images	lesion boxes released; levels pseudo-labelled	corpus (all 10,466 images)
FluoroCap	literature fluoroscopy panels	intraoperative fold banks only, training[f]	not established[g] (source article or video)	950 panels as 2,100 crops, of 29,484 harvested	none released	none: dropped
Real exposures with three-dimensional truth						
C-ArmLSpine	mobile C-arm, Surgio-B51; per-case CBCT	naming cross-validation; held-out detector test	not established; 149 released cases, 148 scored[h] (case = acquisition session)	296 scored: 210 / 30 / 56 (105 / 15 / 28 cases); released 299 spot exposures, two cone-beam sweeps[h]	CBCT, two automatic labellers, agreed levels; pose by registration	corpus; level weights (fold models); no detector weight
DeepFluoro	mobile C-arm, Cios Fusion; cadaver CT	detector training; device parameters	6 specimens (specimen; 4 / 1 / 1)	366 frames: 238 / 24 / 104	CT and C-arm pose per frame	corpus; parameters (40 of 366 frames, R^2 0.905 vs 0.899); detector weights
COBRA2026, Elekta XVI	radiotherapy kV imager; planning CT	detector training; device parameters	145 patients of 467 public Elekta cases[i] (case = patient; 110 / 8 / 27)	3,480 frames: 2,640 / 192 / 648	planning CT, deformably registered; RTK geometry	corpus (all 692 public cases); parameters (40 of 335, R^2 0.974 vs 0.964); detector weights
Geneva hip	mobile C-arm, BV Pulsera; phantom CT	unseen device, test only; device parameters	1 phantom (none: one object)	116 frames, all test	CT and gold-standard pose	corpus; parameters (40 of 116, within-subject R^2 0.952 vs 0.917)
Ljubljana 2D-3D-GS	turntable flat panel, PIXIUM 4600; cadaver CT	device parameters only	1 specimen (not split)	18 frames	CT and gold-standard pose	parameters (14 of 18, within-subject R^2 0.679 vs 0.650); no frame

AP = anteroposterior; CBCT = cone-beam CT; CDC = Centers for Disease Control and Prevention; RTK = reconstruction toolkit; TCIA = The Cancer Imaging Archive; level bank = the training set of the level encoder and head; R^2 = coefficient of determination. [a]Physics-styled re-renders of the three CT collections (7,045 over 3,029 subjects; 1,482 over 81 transitional subjects) repeat subjects already counted; bank subject counts are not additive. [b]Record 0533 was a truncated download and is in no trained bank; rebuilt and replaced in the release (802 records in the released corpus, supplement Table S3), it enters no reported number. [c]No patient identifier in the release: the scan, series or study is the finest recoverable unit; the TotalSegmentator count folds 24 renumbered pairs and three duplicate groups; in the level banks, CTSpinoPelvic1K implant renders were keyed apart from their source CT and, in the first bank, VerSe series as subjects, which put CT-derived persons on both sides of each bank's split (192 in the first, 164 in the second; none from a film corpus), so no CT-render naming accuracy is quoted. [d]3,598 by content: two case pairs share byte-identical films, which the built split merges (supplement Table S3). [e]The CDC index lists 17,075 films; two are served empty at the source, so 17,073 are on disk; one annotated film survives only inside the annotation archive. [f]Fold banks = the capped per-fold crop banks of the intraoperative cross-validation (about 31,400 crops per fold); the film bank behind the plain-film accuracies holds no VinDr or FluoroCap image. The released corpus takes all 10,466 VinDr images and drops FluoroCap (a republished figure is the same patient). [g]A panel keys to its source article or video, which may hold several patients or one patient republished. [h]The archive ships 299 spot exposures over 149 cases (149 AP, 149 lateral with one repeat, one oblique) and two 301-projection cone-beam sweeps; the scored bank holds one AP and one lateral of 148 cases (case134's two, filed in its cone-beam directory, were missed by the export script) and scores case118's −58° oblique as a lateral. No patient tag survives; the cases fall on five acquisition dates. [i]692 public cases: 467 Elekta XVI (centres A–D), 225 Varian OBI (F–G); 152 Elekta cases were fetched before the analysis and 145 kept — two pilot cases (A004, A005) never entered the export list and five (A081, A239, A260, A303, B000) failed an axis-map check, B000 because its planning CT is released outside the imaged volume; 26 Varian cases trained no model; the remaining cases enter no reported number. [j]Totals over the additive rows: 4,093 renders (801 + 1,170 + 1,735 + 387); 9,108 level-bank films (4,963 + 3,603 + 542; 2,071 VinDr films and 950 FluoroCap panels sit in the fold banks alone); 4,276 real frames scored (296 + 366 + 3,480 + 116 + 18), of which the mobile C-arm rows (296, 366, 116) give 778; subjects are not totalled because the units differ. [k]Corpus: rebuilt from the public release by the released exporters (the licences differ, so derived frames are not redistributed). Parameters: a fitted nine-term image-formation file for the device, with the frames it was fitted on and its median R^2 over six folds on the frames each fold left out, against uncalibrated physics (patients left out for Cios Fusion and Elekta XVI; views or frames of the one object for BV Pulsera and PIXIUM 4600). Weights: the released weights were trained on it (level weights = the fold models of the intraoperative cross-validation; detector = the held-out-device detector trained on COBRA2026 and DeepFluoro real frames); corpus and weights are released at publication. Downloaded but used by no trained component: SpinalAI, AASCE, SpineSAC phantom frames, Mendeley-716 and one internal render set.

Table 2: Transitional persons by subtype: definition, persons per source, the truth behind the subtype, the test side of the built person-disjoint split, the share of held-out persons the current level encoder trained on, the unread COBRA2026 screen calls, and the best-case Wilson 95 % lower bound the test count allows.

Subtype[a]	Definition; what a detector can see[b]	Persons: total (CTSP / VerSe / TS / C-Arm)	Truth behind the subtype	Test persons, built split[c]: CT + real	Validation and test persons the current encoder trained on[d]	Unread COBRA2026 screen calls (two-labeller count, with a frame)[e]	Wilson lower bound with no error at the test count, %[f]
L6	six rib-free vertebrae above the sacrum; six lumbar bodies where the sixth carries a label	110 (18 / 86 / 3 / 3)	CTSP two-reader consensus or release label; VerSe published read (38 persons on the audit read alone); TS independent rib count; C-Arm one automatic labeller	43 + 3	49 of 54	38 (3)	91.8 (43 CT); 43.9 (3 real)
T13	thirteen rib-bearing vertebrae	7 (0 / 6 / 1 / 0)	VerSe mask label 28; TS independent rib count	3 + 0	4 of 4	45 (0)	43.9 (3 CT); none (0 real)
Sacralised L5, four lumbar	L5 fused to the sacrum (Castellvi III or IV) and four lumbar labels; the fused body sits in the sacrum label: one box	12 (9 / 3 / 0 / 0)	CTSP two-reader consensus; VerSe published read	5 + 0	14 of 14, both sacralised rows	8 (4)	56.6 (5 CT); 51.0 (4 real)
Sacralised L5, five lumbar	L5 fused to the sacrum on a five-lumbar count; its own label (CTSP) or none (VerSe)	16 (3 / 13 / 0 / 0)	as the row above	6 + 0	with the row above	none called	61.0 (6 CT)
Castellvi I/II	enlarged or articulating transverse process; the count is unchanged	31 (4 / 27 / 0 / 0)	CTSP two-reader consensus; VerSe published read	12 + 0	15 of 15	not screened	75.8 (12 CT)
Lumbar rib	rib on L1; the count is unchanged	14 (14 / 0 / 0 / 0)	CTSP release labels	6 + 0	7 of 7	not screened	61.0 (6 CT)
Unassigned	four lumbar labels, source read normal; no subtype established	4 (4 / 0 / 0 / 0)	CTSP release labels	2 + 0	3 of 3	–	–
L6 and T13 in one spine	none in any truth table	0	–	–	–	–	–
Any subtype	the rows above	194 (52 / 135 / 4 / 3)	–	77 + 3	92 of 97	91 (7)	at-risk rows (L6, T13, four lumbar): 93.0 (51 CT); 64.6 (7 real)

CTSP = CTSpinoPelvic1K; TS = TotalSegmentator; C-Arm = C-ArmLSpine; L5, L6 = fifth, sixth lumbar vertebra; T13 = thirteenth thoracic vertebra. [a]Distinct persons (patient key; two VerSe series pairs are one person each); rib convention: L6 = six rib-free vertebrae between the last rib-bearing vertebra and the sacrum, T13 = thirteen rib-bearing vertebrae. Counted 23 September 2026 from the subtype table; eight of twelve TotalSegmentator spines once called L6 from raw mask identifiers read normal on an independent rib count and are not counted. [b]Whether a detector sees one body or two is decided by the mask: a fused body with its own label gives two boxes, one inside the sacrum label gives one, one left unlabelled gives none. [c]Person-disjoint, subtype-stratified split, 50 / 10 / 40 train / validation / test within each subtype, every real transitional subject on the test side (the three C-ArmLSpine L6); built 23 September 2026, leak audit passed; Figure 5c draws these counts. The three real C-ArmLSpine L6 are scored only in cross-validation, as agreement with one labeller's read (Table 3). [d]Persons on the validation or test side of the built split who sit in the training split of the level bank behind the current encoder (the two most recent training sets); a model descending from those banks cannot score them, so a transitional test needs an arm trained without them. [e]SPINEPS with VERIDAH on the planning CT deformably registered to the cone-beam CT, read by no person; on CTSP this labeller's L6 call matched the readers in 10 of 24 and its T13 call in 0 of 52. The screen also flagged three cases for re-reading. All calls sit on the test side; none counts as a transitional test person until verified. In brackets, counted 24 September 2026, the persons whom this labeller and an independent TotalSegmentator rib count, both requiring the thoracolumbar and lumbosacral junctions in the field, call transitional and who have an exported frame; no T13 can be counted, T1 lying outside nearly every field; four-lumbar persons are placed in the four-lumbar row. [f]Wilson 95 % lower bound when every test person is named correctly, the best case at that count ("real": the bracketed COBRA2026 persons, if each were verified); at risk = L6, T13 and the four-lumbar sacralised L5, the subtypes a count from the sacrum or from C2 misnames; an own-label sacralised L5, Castellvi I/II and a lumbar rib leave the count unchanged and are not in the at-risk bound; the pre-stated targets are 90 % for the exact level (35 persons with no error) and 85 % for variant sensitivity (22 carriers without a miss).

Table 3: Level naming by morphology, counting from the detected sacrum, and the coupled call, per stratum and field size.

Mechanism	Test set; stratum; grouping	View; field[b]; operating point	Measure	% (k/n)	Clusters (with an error)	95 % percentile	BCa
Morphology (level model)	C-ArmLSpine, agreed truth[a]; no C-ArmLSpine frame in training[g]	lateral; whole frame	exact level, full coverage	96.4 (595/617)	141 (6)	Wilson [92.3, 98.4][f]	[92.5, 98.7]
		AP; whole frame	exact level	43.7 (276/631)	142 (83)	[34.9, 51.9]	[34.8, 51.8]
		lateral; three in field	exact level	83.7 (970/1,159)	142 (59)	[79.6, 87.5]	[79.4, 87.4]
		lateral; one in field	exact level	63.8 (385/603)	142 (118)	[59.8, 67.9]	[59.8, 67.7]
	the same; case-grouped folds[g]	lateral; whole frame	exact level	97.4 (601/617)	141 (4)	Wilson [93.2, 99.0][f]	[93.2, 99.4]
		AP; whole frame	exact level	96.4 (608/631)	142 (7)	Wilson [92.5, 98.3][f]	[92.8, 98.4]
		lateral; three in field	exact level	93.1 (1,079/1,159)	142 (23)	[89.2, 96.2]	[88.5, 95.9]
		AP; three in field	exact level	85.5 (1,005/1,176)	143 (63)	[82.2, 88.6]	[82.1, 88.5]
		lateral; one in field	exact level	74.8 (451/603)	142 (94)	[70.9, 78.7]	[70.7, 78.5]
		AP; one in field	exact level	67.5 (399/591)	143 (113)	[64.2, 71.0]	[64.3, 71.1]
	the same; day-grouped folds[g]	lateral; whole frame	exact level	92.1 (568/617)	141 (12)	[87.5, 96.0]	[86.7, 95.6]
		AP; whole frame	exact level	94.8 (598/631)	142 (9)	Wilson [90.1, 97.3][f]	[90.2, 97.5]
		lateral; three in field	exact level	87.7 (1,016/1,159)	142 (38)	[83.5, 91.6]	[82.5, 91.1]
		AP; three in field	exact level	86.0 (1,011/1,176)	143 (63)	[82.6, 89.1]	[82.3, 88.9]
Count from the detected sacrum, no classifier[c]	C-ArmLSpine test, reported detector	whole frame; conf 0.74	counting defined (a sacrum detected)	82.1 (46/56)	28	[69.6, 92.9]	[69.6, 91.1]
			ordinal correct, matched bodies	95.4 (188/197)	28	[89.4, 99.5]	[85.2, 98.9]
			truth bodies at the right ordinal, defined frames	81.0 (188/232)	28	[73.8, 87.6]	[73.5, 87.1]
		whole frame; conf 0.25	ordinal correct, matched bodies	84.7 (199/235)	28	[77.4, 90.7]	[76.5, 90.4]
	COBRA2026 test, held-out detector	conf 0.25	ordinal correct, matched bodies	71.1 (54/76)	27	[56.6, 80.3]	[57.7, 80.6]
	pooled over the three held-out devices	conf 0.25	ordinal correct, matched bodies	81.0 (306/378)	56	[75.7, 85.8]	[75.7, 85.8]
Coupled: classifier and count[d]	C-ArmLSpine anchored frames, agreed truth	whole frame; conf 0.74	count alone, counted bodies	97.0 (979/1,009)	130	[94.2, 99.2]	–
			classifier alone, counted bodies	97.7 (986/1,009)	130	[95.9, 99.2]	–
			coupled, counted bodies	98.2 (991/1,009)	130	[96.5, 99.5]	–
			coupled minus count; minus classifier (points)	+1.2; +0.5	130	[0.0, 3.2]; [−1.5, 2.4]	–
			disagreement; stop (13 disagreement, 10 low margin); erring caught	5.6 (13/233); 9.9 (23/233); 5/6	132	[2.6, 9.2]; [5.7, 14.5]	–
			L6 calls on five-lumbar frames, every one stopped	3/233	2	no interval	–
		whole frame; conf 0.25	count alone, counted bodies	96.1 (1,104/1,149)	139	[93.5, 98.3]	–
			classifier alone; coupled, counted bodies	97.7 (1,122/1,149); 97.1 (1,116/1,149)	139	[95.8, 99.2]; [95.3, 98.8]	–
			disagreement frames	7.1 (19/266)	142	[3.8, 10.8]	–
Transitional, real exposures	three C-ArmLSpine cases read as L6 by one labeller[e] (Table 2)	whole frame; conf 0.25 and 0.74	agreement with that read on the L6 body (coupled; classifier); anchored frames called L6 (conf 0.74)	1/5; 0/5; 1 of 4	3	no interval	–

AP = anteroposterior; BCa = bias-corrected and accelerated bootstrap interval; conf = detector confidence threshold. Intervals are 95 % bootstraps of 2,000 resamples clustered by case (intraoperative) or patient (film); "–" = not computed. [a]Agreed truth = the levels both automatic CBCT labellers name (814 of 910 vertebrae; T11 and most of T12 withheld); ordinary spines. Case-grouped = five folds by case, day-grouped = three folds by acquisition day (a case is an acquisition session, not a proven person); "with an error" = clusters carrying at least one error. [b]Field: whole frame, or crops holding three or one vertebra. Exact level scores every vertebra the film-trained keypoint detector boxed (confidence 0.35; 617 of 676 agreed lateral; with the 59 undetected counted wrong, 88.0 % zero-shot) on its decoded name at full coverage; naming matches a detection to the agreed vertebral-body box containing its centre (overlap at least 0.3, one to one). [c]Ordinal = rank of the detected body away from the detected sacrum, scored name-free against the rank from the truth sacrum; defined when truth has a sacrum and one was detected; one-to-one matching at intersection over union ≥ 0.5, whole-vertebra convention (supplement, Table S2); a missed truth body counts as wrong in the defined-frames row. C-ArmLSpine test = 28 cases, 56 AP and lateral frames, device never seen; COBRA2026 test = 27 patients, 609 frames with a sacrum, detector trained on DeepFluoro and C-ArmLSpine real frames; pooled = each real-frame detector on its unseen device. [d]C-ArmLSpine five-fold CV frames with agreed truth (145 cases, 290 frames), the sacrum from the held-out detector. Classifier = the level model's name (the case-grouped fine-tuned fold models; AP and lateral frames pooled); count = the ordinary alphabet at the ordinal (L5 at one), no classifier; coupled = the alphabet (ordinary, L6, T13, both) that best fits the classifier's posteriors at the ordinals; the read stops when classifier and call disagree or the winning alphabet's margin over the runner-up is below the fifth percentile of correct frames in the other four folds; the level model abstains on a body when its decoded and single-crop names disagree or its posterior is below 0.5, a default never tuned; percentile intervals only. [e]Agreement with that read, not correctness: the two labellers segment the same five bodies and disagree only on the names (L1–L5 against L2–L6) on a lumbar-only CBCT; six frames, three clusters, no interval. [f]Wilson 95 % interval on the design-effect-adjusted count where the result file flags errors in few clusters. [g]Zero-shot = the encoder and its recurrent head with no C-ArmLSpine case in training (checked for all 148 cases); the fold models are fine-tuned from that encoder, every fold held out by list intersection.

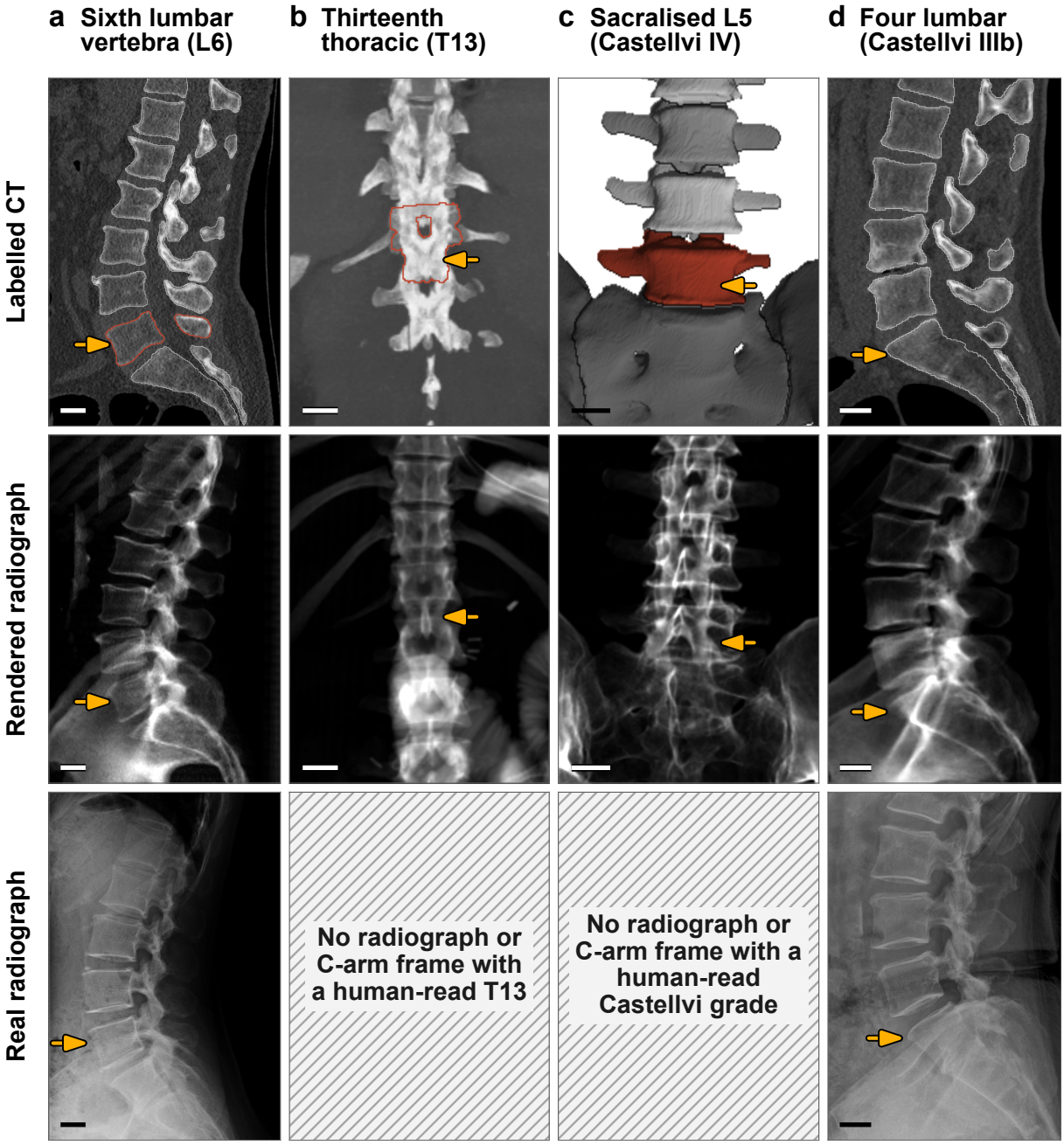


Figure 1: Transitional phenotypes. (a) Sixth lumbar, 54-year-old man; (b) thirteenth thoracic, 24-year-old woman; (c) sacralised L5, own label, Castellvi IV, 89-year-old, sex other; (d) four lumbar, L5 in sacrum, Castellvi IIIb, 60-year-old man. Rows: labelled CT (sagittal; b, coronal slab; c, surface); render; another patient's film (women, 31, 64). Arrow and red, transitional vertebra; bar, 20 mm; hatched, none human-read.

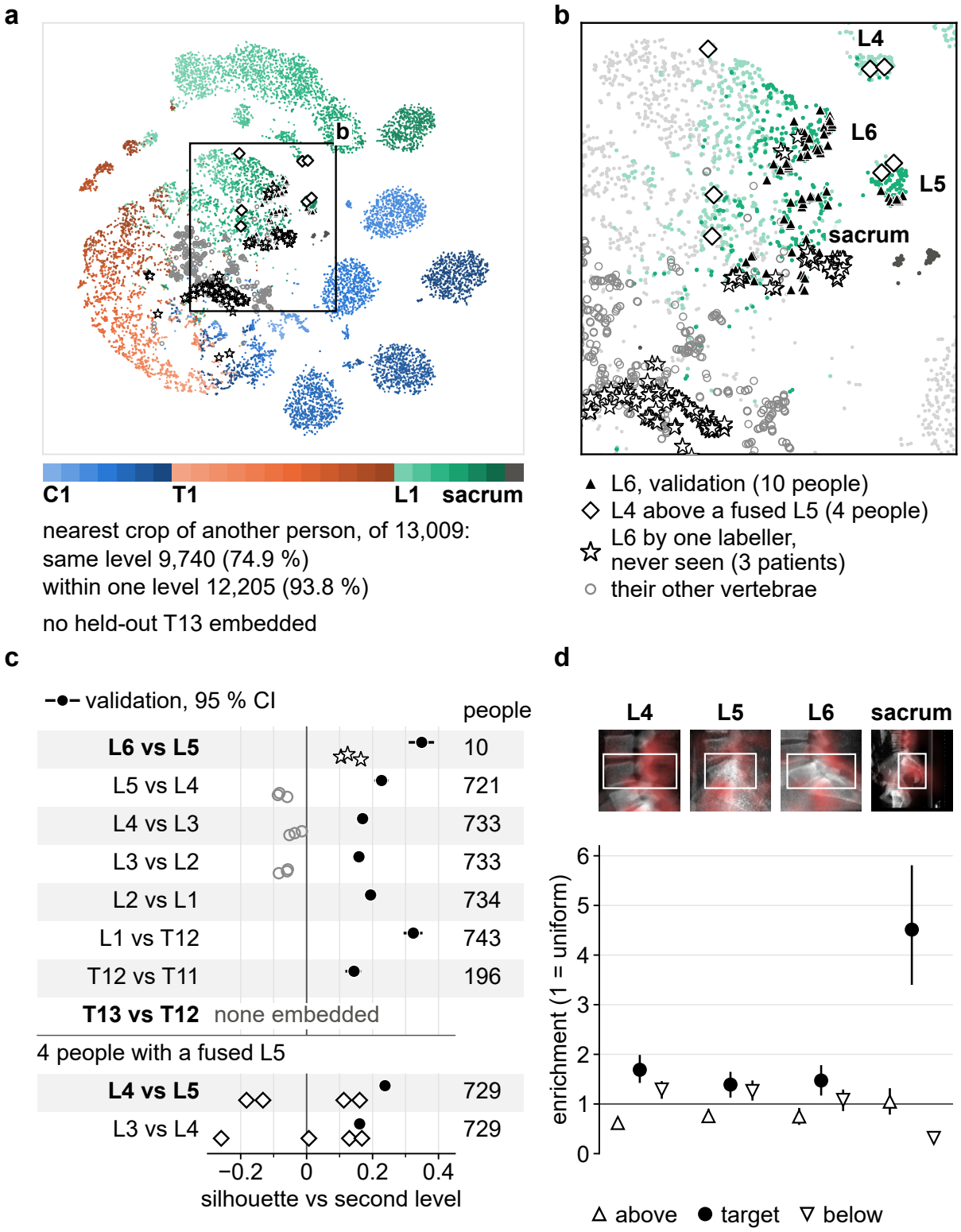


Figure 2: Level-encoder embedding, validation crops (checkpoint-selection split). (a) t-SNE (t-distributed stochastic neighbour embedding) by level; triangles, sixth lumbar vertebra (L6); diamonds, last free vertebra above fused L5; stars, one labeller's L6 in never-seen patients; rings, their other vertebrae; T13, thirteenth thoracic. (b) Lumbosacral window. (c) Silhouette; CI, confidence interval. (d) Occlusion enrichment: attribution over area share; ten lateral crops per level.

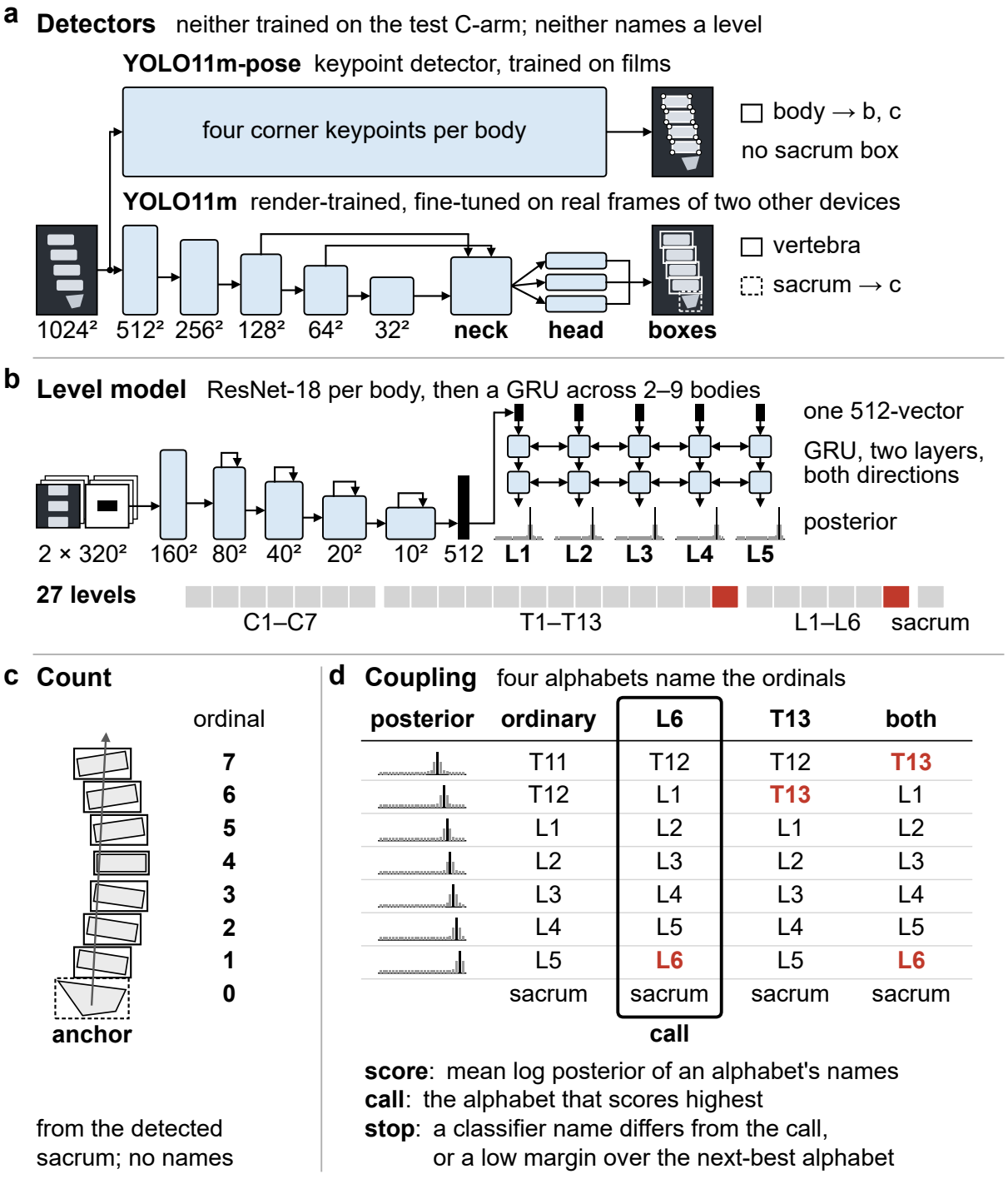
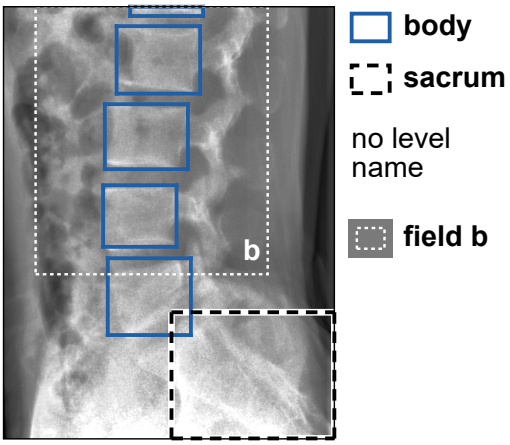


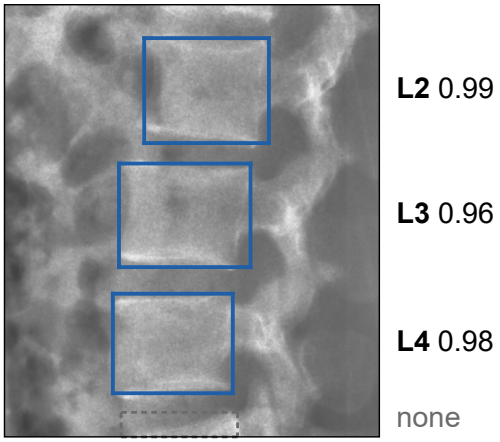
Figure 3: SchwingNet; shaded blocks learned, sizes in pixels. (a) Two detectors, neither naming a level; one boxes bodies, one the sacrum. (b) Level model: two-channel crop per body, ResNet-18, bidirectional gated recurrent unit (GRU) across bodies, 27-level posterior; red, T13 (thirteenth thoracic) and L6 (sixth lumbar). (c) Ordinals counted from the detected sacrum. (d) Posteriors choose the alphabet; outlined, the call.

a detect



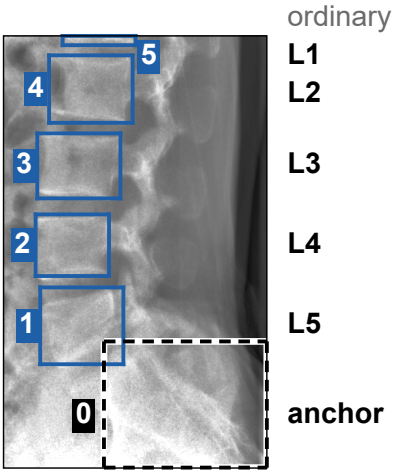
Real lateral C-arm exposure; neither detector was trained on this C-arm. Lateral bodies boxed: 617 of 676 (91.3 %); sacrum at confidence 0.74: 233 of 290 frames, both views (80.3 %).

b name from shape alone



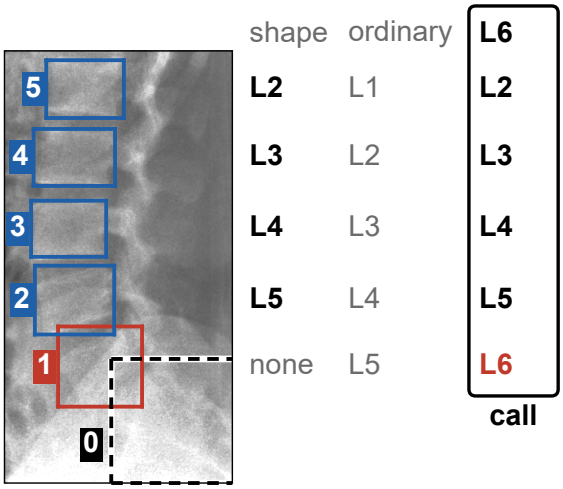
Neither end of the spine is in view: nothing to count from. Each body is named by its shape (posterior beside), by a model fine-tuned on other patients. Three in view: 1,079 of 1,159 bodies exact (93.1 %).

c count from the sacrum



Same frame. Ordinals from the detected sacrum (0), named in the ordinary, five-lumbar alphabet; no shape is used. Count alone, sacrum at 0.74: 979 of 1,009 bodies exact (97.0 %).

d six lumbar: both must agree



Shape fits L6 on 4 of 4 named bodies, ordinary on 0: call L6; body 1 unnamed: stop. CT labellers: L2–L6, L1–L5. L6 calls, all stopped: 1 of 4 anchored frames of three patients one labeller reads six-lumbar; 3 of 233 both read five-lumbar.

Figure 4: Reading C-arm exposures. (a) Bodies (blue) and sacrum (dashed) boxed by detectors never trained on this C-arm. (b) Field b, enlarged, named by shape, fine-tuned on other cases ("none": no name). (c) Ordinals from the sacrum, ordinary alphabet. (d) A frame one CT labeller reads six-lumbar: shape fits the L6 alphabet (outlined); a stop, not a verdict. L1–L6, lumbar vertebrae.

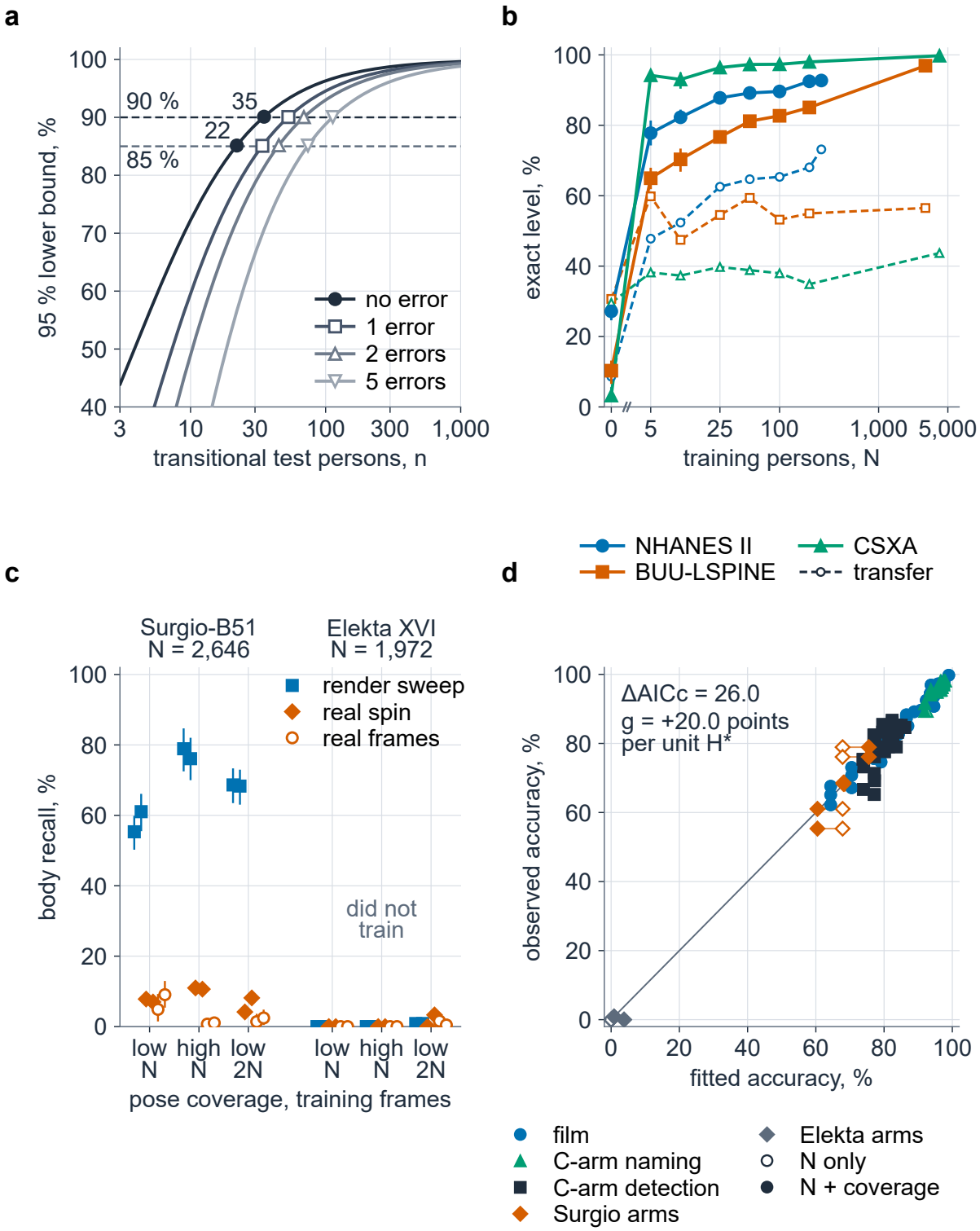


Figure 5: Persons, films, coverage. (a) Wilson 95 % lower bound by errors against transitional test persons; lines, targets. (b) Naming accuracy against training persons per film corpus; bars, seed range; dashed, tested on NHANES II (NHANES II: on BUU-LSPINE). (c) Render-only detectors at matched N, low- against high-coverage poses. (d) Observed against fitted accuracy: N only (open), N plus coverage (filled).

Supplemental material

Supplemental material for *SchwingNet: a convolutional-recurrent network for level naming in spine surgery*. Figures S1–S5 (flow of cases; corpus manifold; image-formation parameters; which training source carries the detector; detection against view angle) and Tables S1–S3 (image-formation terms per device; detection per held-out device and training arm; corpus by source, with age and sex) are numbered by first citation in the main text; the last section documents the evaluation gate and every audit tool the study added. Every count is read from the file named in the repository's NUMBERS.md.

Corpus manifold

A classifier telling films from fluoroscopy with one corpus held out (19 corpora, 7,311 frames) differed from the majority-class share of 71.8 % by -0.7 to 9.2 points across three feature spaces and two classifiers, no percentile interval excluding zero (two nearest-neighbour bias-corrected and accelerated intervals did). The probe drawn in Figure S2b is a different one, films against C-arm and kilovoltage-imager frames with one source held out (5,797 frames): on image statistics it read 63.4 % (57.6, 69.2) against a majority share of 64.4 %, and on deep features 85.0 % (79.4, 90.0), so on those features the modalities are separable. Each device's styled renders remained separable from its real frames on the same features (Figure S2c): 100 % (Cios Fusion, 736 frames), 99.8 % (Surgio-B51, 666) and 99.0 % (Elekta XVI, 771; 98.1, 99.6), against majority shares of 50.3–64.7 %. Detectors are therefore held out by device, and no claim rests on the modalities being indistinguishable.

Image-formation terms

Renders were styled through nine image-formation terms fitted per device, leave one group out, on up to 40 real frames (14 of 18 for the PIXIUM 4600) against uncalibrated physics (Figure S3, Table S1). On the two devices whose folds hold out patients (Cios Fusion and Elekta XVI; 12 folds, 701 frames), the fitted terms raised held-out coefficient of determination (R^2) by 0.019 (95 % confidence interval: 0.008, 0.033; frame-weighted 0.023; 0.010, 0.033), two folds negative (exact sign-flip over folds $P = .004$, Holm-adjusted $P = .009$). On the other two a fold holds out views or frames of the one object every fold was fitted on (the BV Pulsera phantom, the PIXIUM 4600 cadaver), so their gain, 0.056 (0.025, 0.088; 12 folds), is within-subject, not held out. The pre-stated criterion, a gain above zero with the interval excluding zero on every device, was not met: it held only on six-fold percentile bootstraps, and on six-fold t intervals only the BV Pulsera gain excluded zero (Cios Fusion 0.015; -0.003, 0.034). A refit with image noise in the loss failed its own acceptance gate on every device it was run on, and the published parameters stand. Exposure draws were clipped to the central 60 % of each marginal, a defect not repaired.

Which source carries the detector

Transfer to the held-out C-arm, scored against whole-vertebra boxes (the convention of the training sources), tracked the source whose frames span the gantry arc (Figure S4): the radiotherapy imager alone

read 86.8 % (249 of 287) of held-out vertebrae, the cadaver C-arm alone 12.9 % (37) (against vertebral-body boxes, a box-convention reading, 11.1 % and 3.8 %). Adding the wide-arc source raised held-out recall by 71.8 points (63.4, 79.6; second training seed 67.6; 57.6, 76.8), adding the cadaver source changed it by -2.1 (-6.0, 1.4; second seed -6.3; -9.8, -2.9), and cutting the wide-arc frames from 2,640 to 240 dropped it by 47.7 points (41.8, 53.7; body boxes, -8.0, no paired interval). At the default confidence the same-device control found the sacrum on 55 of 56 frames (98.2 %), 14.3 points more than the held-out detector (5.4, 23.2; Holm-adjusted exact sign test $P = .009$). The sources differ in device, anatomy, frame count and angle together, and no arm isolates angle. Every detector trained on real frames alone, the reported one included, was fine-tuned from the render-trained detector. Renders alone found 31.7 % (91 of 287) at the default confidence (body boxes, 2.4 %) and one vertebra at 0.74; on 161 vertebral bodies of 17 real films whose patients neither detector trained or was selected on, a film-trained detector found 31.7 points more than a render-trained one at confidence 0.05 (99.4 % against 67.7 %; 21.0, 45.4; 16 of 17 films favouring the film-trained detector, none the other; Holm-adjusted $P < .001$).

Plain-film naming on the selection split

On the patient-keyed validation split that also chose the encoder checkpoint (one seed), with random windows of two to nine bodies, the level model named 3,629 of 3,635 CSXA cervical vertebrae (99.8 %; 727 patients, three with an error; Wilson on the design-effect-adjusted count 99.5, 99.9; BCa 99.5, 99.9), 2,671 of 2,744 BUU-LSPINE lumbar vertebrae (97.3 %; 549 cases, 51; percentile 96.5, 98.0; BCa 96.4, 98.0) and 441 of 445 NHANES II lateral vertebrae (99.1 %; 76 films of 52 patients, four; Wilson 97.8, 99.6; BCa 98.0, 99.8); with three bodies in view, 99.6 %, 95.0 % and 94.6 %. Every annotated patient of the three corpora is in this encoder's training or selection split, so no held-out plain-film accuracy exists until a model trained from ImageNet weights on the split's training persons is scored on the test films of the person-disjoint split, which is not done here.

Same-device frames and view angle

On the C-ArmLSpine test split (28 cases, 56 frames, 287 vertebral bodies on the repaired truth; body boxes, unlike Table S2), a vertebra detector whose training held no frame of the device found 7.0 % (20 of 287; 95 % case-bootstrap interval 4.2, 10.1) at confidence 0.74 and 12.5 % (36 of 287; 9.0, 16.1) before its own-geometry renders were added. With 50 real frames of 25 cases it found 82.9 % (238 of 287; 78.5, 87.2); with 100 frames 83.3 % and 81.9 % (two seeds); with 200, 81.5 % and 83.6 %; with all 210 frames, 85.7 % (real frames only, from public weights), 85.0 %, 83.3 % and 86.8 % (249 of 287; 82.3, 90.9) for the arms adding own-geometry renders, the other C-arms' frames, or both. Every interval from 50 frames upward overlaps every other; at the default confidence the arms read 89.5–92.3 %. The subsets are nested and drawn by whole case, no test case or patient is in any list, and the styled renders of this device were posed on the registered geometry of all 148 cases, the 28 test cases included. The real pose pool of the device holds 148 anteroposterior (within 10°) and 148 lateral exposures and no pose between: on an acquisition-space grid of 18 lateral-angle bins of 10° and six tilt bins, the registered poses occupy three lateral bins (normalised entropy 0.19) and 6 % of the joint bins, against 83 % of the lateral bins for the one rotational acquisition. The learning-curve training sets occupy every lateral bin through their other-device frames and renders, and their joint coverage (37 %) and entropy (0.63) are the same at 50, 100 and 200 C-ArmLSpine frames, so the added frames re-sampled the same bins and count was never separated from coverage; no arc pose entered any training list. For the angle sweep, render pose pools were extended per device to an arc from $-90°$ to $+90°$ (592, 1,100 and 4,216 poses; tilt within about $\pm 13°$, 5th to 95th percentile). On 40 TotalSegmentator v3 subjects that no scored detector trained or was selected on, the pooled detector's body recall at the default confidence fell below 90 % of its anteroposterior value by $\pm 30°$ on both device cards, dipped to 0.69 at $-50°$ on the Surgio card and recovered to 0.99 at the

lateral; because every subject is placed on one anchor, another subject's truth boxes already score 0.64–0.68 at every angle, so those values are read against a floor of about 0.67, not zero, and the Surgio lateral is rendered photon-starved by its card. On one patient's real rotational acquisition, recall against view angle is drawn in Figure S5; one patient admits no interval, and the reported detector's flat recall there is unexplained.

Development learning curve of a split-trained level model

The recurrent head of a level model trained from ImageNet weights on the split's training persons (the arm a transitional test needs, Table 2 note d of the main text) was also trained on nested fractions of those persons, 563, 1,121, 2,240, 4,477 and 8,950, three seeds each, and read on the validation side that chose its epoch; these are development numbers, and no test person is among them. On random windows of two to nine bodies, accuracy rose from 68.9 % to 74.0 % across the sixteen-fold range, the last doubling adding 0.6 points against a spread of 1.0 across seeds at full size. On real films it was 98.2 % at 563 persons and 98.7 % at 8,950 (last doubling 0.1 points; with three bodies in view, 97.2–97.5 % at every size); on CT renders 59.6 % to 66.2 % (0.8); on rendered L6 bodies 51.8 % to 64.2 % (1.7 points, seed spread 3.4); rendered T13 bodies were named in none of the three full-size runs. The 30-epoch runs came within 0.5 points of their final validation accuracy by the 8th to 12th epoch (three seeds) and moved within 0.3 points over the last five. The bank records no projection angle, source distance or magnification, so its acquisition-space coverage can be counted only over source, view and pose index; five lateral pose cells of the transitional CTSpinoPelvic1K renders occur on the validation side and not in training.

Transitional crops in the embedding

Within single render sources of the checkpoint-selection split, the silhouette of L6 from L5 exceeded the L5-from-L4 control in two, the VerSe renders (0.52; 95 % confidence interval: 0.45, 0.57; against 0.38; ten persons) and the transitional-enriched physics renders (0.24; 0.20, 0.29; against 0.16; eight persons), and fell below it in the third, the standard physics renders (0.14; 0.11, 0.20; against 0.21; five persons, 11 crops).

Three C-ArmLSpine patients whose cone-beam CT one automatic labeller reads with a sixth lumbar vertebra, and the other as L1–L5, are in neither split of the level encoder; their crops were rendered under six device cards and embedded (Figure 2c of the main text, stars). Under the pre-specified whole-vertebra box, the body read as L6 sat nearer the validation L6 than the L5 cluster in each patient (two-cluster silhouette 0.10, 0.12 and 0.16; validation L6 left out by person, 0.34), their L3–L5 did not (-0.09 to -0.01), and 84 of its 180 crops were named L6. Under the secondary vertebral-body box the result vanished (-0.07; 13 of 180 named L6). The negative L3–L5 values are what the second reading (L2–L4) predicts, so apart from the disputed body the panel does not decide between the readings; the crops are three patients, not 180 observations.

Four validation persons with a fused L5 left unlabelled (seven crops of the last free vertebra, which a count up from the sacrum names L5) were placed between the validation L4 and L5 clusters. The last free vertebra sat nearer L5 than L4 in two of them, one crop each (-0.13 and -0.18; ordinary L4, 0.24, with 70 of 1,066 crops below zero), and both crops were named L5, the name a count gives; the second of the two also had its L3 named L4. Their L3, the control, was also low (0.01 and -0.26; ordinary L3, 0.16), so four persons cannot separate a junction effect from a person-level shift.

Corpus census

The corpus as built — persons and primary images per source and split side, transitional persons, and age and sex where a release gives them — is Table S3, computed from the person-disjoint split of 23 September 2026, the transitional registry and each release's metadata (tools/corpus_census.py, tools/age_sex_by_source.py). Where it differs from Table 1 of the main text the two count different objects: Table 1 counts what each trained or scored component consumed (bank, fold-bank crops, the detector's earlier case splits), the census what the split places. So CTSpinoPelvic1K record 0533, rebuilt after the banks were frozen, is a person and a render in the census (802) and in no trained bank (801); the split folds TotalSegmentator scans by content hash to 1,867 groups against Table 1's 1,935 scans, and BUU-LSPINE's two byte-identical case pairs to 3,598 persons; NHANES II, VinDr-SpineXR and COBRA2026 enter the split whole (692 COBRA2026 cases, 145 with an exported frame) where Table 1 counts the 364 annotated persons, the 2,071 fold-bank films and the 145 detector cases; the 3,600 BUU-LSPINE anteroposterior films are placed by the split and were in no level bank; and the real-frame sides are the split's person-hash draw, not the detector's fixed case splits that Table 1 prints. Every printed cell of Table 2 reproduces from the subtype table it was drawn from; the registry of 24 September adds labeller-tier and read-tier calls no table counts until verified (TotalSegmentator 22 L6 and 26 unassigned, COBRA2026 three L6 and six unassigned, and on VerSe 12 lumbar ribs and 20 unassigned on the read tier; two VerSe L6 screened negative). One count remains unexplained: footnote a of Table 1 gives 7,045 physics-styled renders against 7,048 items in that root; the 18 Ljubljana frames and the second released C-ArmLSpine sweep are in no split root and are counted from Table 1 alone.

Data integrity and quality controls

Each control was added after a defect was found and is implemented in the named file of the released code; a number a control removed was replaced or marked pending, never left. **Completeness audit:** every source release was audited against what is on disk and what each model consumed; of 99 gaps, 31 were held back by a written criterion, 28 came from software defects, 21 from unfinished jobs, four from download or compute cuts with no criterion and 15 from causes nobody can establish, and every gap touching a quoted number is footnoted in the tables (a truncated CT record, two films served empty at their source, one mislabelled oblique exposure, two never-built intraoperative frames); the audit ships with the released corpus recipe. **Patient-keyed evaluation gate:** `tools/eval_gate.py` runs before every quoted score and raises unless each item's patient key is absent from the training and checkpoint-selection persons of the checkpoint and of every checkpoint it was initialised from (public ImageNet or COCO weights the only exempt ancestor; a render inherits its CT subject; an unreadable link is an error, not a pass), stamps the verdict and the weights' SHA-256 into the result file and refuses a file whose stamp does not match; a bank that cannot be found raises instead of returning an empty seen set (`tools/holdout.py`). **Leak check CHK-9** of `tools/leak_check.py`, over the 563 numbers then quoted (`results/eval_gate/quoted_results.json`), found 323 held out, 81 scored on a seen patient and 58 with an unreadable lineage (101 involve no checkpoint); each of the 81 and 58 was re-derived on unseen persons, replaced or marked pending (`results/eval_gate/EVAL_LEAKS.md`). **Checkpoint lineage:** `tools/checkpoint_lineage.py` reads the stored training configuration of each of 306 checkpoints and writes the persons seen along its chain; 133 carry a link that cannot be read and certify nothing (`results/eval_gate/LINEAGE.md`); every quoted zero-shot and fine-tuned number comes from an encoder whose whole chain was read (Table 3, note g). **Patient keys and split builder:** `tools/make_split.py` keys every item to one person, sends synthetic, warped and implant variants with their source patient, allocates transitional persons 50:10:40 per subtype with every real transitional subject on the test side, and runs checks CHK-1 to CHK-8 (item and person disjointness under two keyings, alias and derivation closure, real transitional subjects held out whole, near-duplicate images across

sides, undeclared aliases, every item keyed), exiting non-zero on a failure and writing a check that did not run as not run; its cross-source patient key came from a defect `tools/leak_check.py` found, one radiograph collection entered twice under two naming schemes with 48 of 54 of its validation patients twinned in training.

Label-identifier guard: `tools/spineps_ids.py` raises unless a segmentation tool's instance identifiers lie in the VerSe alphabet, run cranio-caudally in world coordinates and agree with the ordered names written beside the mask, and every reader that turns an identifier into a level calls it; 2,437 harvested masks carried raw identifiers 1 to N, copied by 2,714 body masks, which read as names had selected twelve TotalSegmentator spines as six-lumbar (an independent rib count confirmed four, refuted eight), and the 3,490 masks were relabelled into new trees beside the old (`results/spineps_ids/`). **Results-driven build:** the numbers keyed in `numbers_manifest.json` are read at compile time from the results files it names (`tools/results_macros.py`), and a missing result prints as a pending marker, not a typed number. **Box convention:** all 69 quoted detector numbers were re-scored in both conventions, whole-vertebra outline and vertebral body, which cross on every device, so each printed detector number names its convention (`results/bodybox_rescore/RESCORE.md`). **Excluded sources:** `tools/excluded_sources.py` raises when a path from an excluded collection reaches a list, bank or probe (the Mendeley lumbar set, 500 of whose 584 films re-encode BUU-LSPINE films already in training; the literature fluoroscopy panels, figures of published articles; Table 1). **Release repairs:** CT-SpinoPelvic1K record 0533 was a truncated download (17,301,504 of 250,067,995 bytes), rebuilt from the archive, verified by nine checks and replaced in the public release, and enters no reported number (`results/repairs/ct0533_CHECK.md`); one detector arm had counted COBRA2026 frames up to four times through overlapping list roots (5,013 duplicate entries), and the bank registry now refuses an unregistered directory and checks split disjointness on disk (`tools/realxray_train.py`; `results/repairs/cobra-dup-leak_CHECK.md`).

Table S1: The nine image-formation terms fitted per device: median over the leave-one-group-out folds (fold minimum–maximum), with the unit and the fit bounds. Released as one parameter file per device.

Term	Meaning	Unit	Bounds	BV Pulsera	Cios Fusion	Elekta XVI	PIXIUM 4600
kb	bone-path scale	–	0.5–2.5	2.5 (2.35–2.5)	1.97 (1.92–2.15)	0.5 (0.5–0.758)	2.5 (2.17–2.5)
s	scatter share	–	0–0.75	0.606 (0.567–0.634)	0.238 (0.197–0.396)	0.245 (0.202–0.311)	0.661 (0.533–0.687)
σ_s	scatter blur	px at fit pitch	5.76–16	16 (16–16)	16 (15.8–16)	15.4 (15–15.6)	14.6 (14.6–14.6)
σ_r	detector blur	px at fit pitch	0.256–3.07	0.269 (0.256–0.333)	0.866 (0.662–0.917)	0.754 (0.739–0.797)	0.233 (0.233–0.315)
α	edge-enhancement (unsharp-mask) gain	–	0–2	0.83 (0.589–1.15)	0.2 (0.111–0.38)	0.287 (0.0525–0.32)	1.85 (1.02–2)
σ_u	edge-enhancement radius	px at fit pitch	0.961–7.94	3.22 (2.79–3.49)	2.89 (2.78–4.65)	2.85 (2.35–2.92)	3.66 (3.34–5.16)
b	radial vignette	–	-0.6 to 0.6	-0.358 (-0.373 to -0.282)	-0.0786 (-0.102 to -0.037)	-0.0339 (-0.0461 to 0.0175)	0.6 (0.6–0.6)
c	anode-heel gradient	–	-0.3 to 0.3	0.017 (0.00663–0.0519)	0.0488 (0.0265–0.0814)	-0.3 (-0.3 to -0.286)	0.0322 (-0.11 to 0.3)
d	cranio-caudal tilt gradient	–	-0.3 to 0.3	-0.145 (-0.149 to -0.137)	-0.0382 (-0.0607 to -0.0227)	-0.0188 (-0.0589 to 0.0228)	0.00423 (-0.0218 to 0.189)
	frames fitted; fold-median R^2, card vs physics			40 of 116; within-subject R^2 0.952 vs 0.917	40 of 366; held-out R^2 0.905 vs 0.899	40 of 335; held-out R^2 0.974 vs 0.964	14 of 18; within-subject R^2 0.679 vs 0.65

R^2 is the median over six folds of the frames each fold left out, card against the uncalibrated physics default; a Cios Fusion or Elekta XVI fold leaves out patients (held out), a BV Pulsera (one phantom) or PIXIUM 4600 (one cadaver) fold leaves out views or frames of the object the other folds were fitted on (within-subject); px = pixels at the fit pitch. Sources: the released device parameter files.

Table S2: Detection per held-out device and training arm at the cross-device (0.74) and default (0.25) operating points. Whole-vertebra box convention throughout; the all-device detector trained on vertebral-body boxes is reported in the separate held-out study.

Target device[a]	Training arm	Vertebra recall at 0.74[b]	Sacrum found at 0.74	Vertebra recall at 0.25	Sacrum found at 0.25	PR-AUC[c], vertebra / sacrum
C-ArmLSpine (Surgio-B51)	Reported detector: real COBRA2026 and DeepFluoro frames, no render	80.5 (231/287) [74.1, 86.8]	76.8 (43/56) [64.3, 89.3]	90.6 (260/287) [84.3, 94.5][g]	83.9 (47/56) [73.2, 92.9]	–
	Design B: styled renders and real frames of the two other devices; seed 1	63.1 (181/287) [54.8, 71.0]	66.1 (37/56) [53.6, 78.6]	87.8 (252/287) [81.4, 93.2]	85.7 (48/56) [76.8, 92.9]	0.890 / 0.900
	Elekta XVI only, styled and real	–	–	86.8 (249/287) [81.1, 91.8]	78.6 (44/56) [67.9, 89.3]	–
	Cios Fusion only, styled and real	–	–	12.9 (37/287) [8.3, 18.1]	AP 28/28, lateral 0/28[g]	–
	Elekta XVI only, real frames cut to 240 in 10 patients	–	–	39.0 (112/287) [33.2, 44.7]	37.5 (21/56) [28.6, 46.4]	–
	Renders only, no real frame	0.3 (1/287) [0.1, 1.9][g]	5.4 (3/56) [1.8, 14.6][g]	31.7 (91/287) [27.1, 36.1]	53.6 (30/56) [44.6, 62.5]	–
	Control[d]: C-ArmLSpine real training frames only	40.4 (116/287) [37.8, 42.9]	94.6 (53/56) [85.4, 98.2][g]	41.5 (119/287) [38.8, 44.0]	98.2 (55/56) [90.6, 99.7][g]	0.367 / 0.979
	Control[d]: real training frames of all three devices	40.1 (115/287) [37.7, 42.3]	96.4 (54/56) [87.9, 99.0][g]	40.8 (117/287) [38.4, 43.0]	98.2 (55/56) [90.6, 99.7][g]	0.385 / 0.982
COBRA2026 (Elekta XVI)	Real DeepFluoro and C-ArmLSpine frames, no render	6.2 (55/893) [2.9, 9.4]	3.4 (21/609) [1.3, 5.9]	18.5 (165/893) [13.7, 22.8]	13.3 (81/609) [8.1, 18.8]	–
	Design B: styled renders and real frames of the two other devices	10.1 (90/893) [6.7, 13.1]	13.0 (79/609) [9.0, 17.1]	32.7 (292/893) [28.4, 36.2]	36.5 (222/609) [31.6, 40.9]	0.266 / 0.391
	Surgio-B51 only, styled and real; two runs of one list	–	–	7.5 (67/893) [3.5, 12.1]; 12.2 (109/893) [8.0, 16.0]	3.8 (23/609) [1.8, 6.2]; 8.0 (49/609) [5.3, 10.9]	–
	Control[d]: real training frames of all three devices	93.2 (832/893) [86.4, 97.7]	90.6 (552/609) [83.8, 96.3]	97.4 (870/893) [93.3, 99.8]	97.9 (596/609) [95.7, 99.5]	0.963 / 0.985
DeepFluoro (Cios Fusion)[e]	Real COBRA2026 and C-ArmLSpine frames, no render; test specimen	74.7 (62/83)[e]	100.0 (90/90)[e]	95.2 (79/83)[e]	100.0 (90/90)[e]	–
	Design B: styled renders and real frames of the two other devices; six specimens	–	–	78.6 (338/430) [68.8, 92.0]	95.0 (321/338) [78.5, 99.8]	–
	The same arm; test specimen	69.9 (58/83)[e]	87.8 (79/90)[e]	94.0 (78/83)[e]	100.0 (90/90)[e]	0.923 / 0.954
	Control[d]: real training frames of all three devices; test specimen	73.5 (61/83)[e]	100.0 (90/90)[e]	95.2 (79/83)[e]	100.0 (90/90)[e]	0.685 / 0.968
Plain radiographs[f]	Render-trained detector, no real image	at 0.05: 67.7 (109/161)	–	34.8 (56/161)	–	–
	Film-trained detector	at 0.05: 99.4 (160/161)	–	98.8 (159/161)	–	–

Cells are % (k/n) [95 % CI]. CI = confidence interval; PR-AUC = area under the precision–recall curve; IoU = intersection over union. Greedy one-to-one matching at IoU ≥ 0.5; 95 % percentile bootstraps of 2,000 resamples clustered by case (C-ArmLSpine), patient (COBRA2026) or specimen (DeepFluoro), BCa intervals in the Markdown twin; "–" = not scored at that point or not computed. Styled = rendered through the fitted terms; design B = renders under each device's own poses. [a]Test sets: C-ArmLSpine 56 frames, 28 cases, 287 vertebrae, 56 sacra; COBRA2026 648 frames, 27 patients, 893 vertebrae, 609 frames with a sacrum; DeepFluoro six specimens, 366 frames, 430 vertebrae, 338 with a sacrum; plain radiographs 17 NHANES II lumbar films (17 spines, 161 bodies) whose patients neither detector trained or was selected on. Truth boxes: C-ArmLSpine rows against whole-vertebra outline boxes projected from the CBCT labels, the convention every number on disk uses (under the SPINEPS vertebral-body boxes of the pending flagship rows the reported detector reads 5.2 % (15/287) at 0.25, a convention mismatch); COBRA2026 and DeepFluoro rows against whole-vertebra boxes projected from the registered CT. The device is held out unless the row says control. [b]0.74 = best pooled vertebra F1 of the reported detector's sweeps on COBRA2026, DeepFluoro and the Geneva phantom, reading no C-ArmLSpine frame, adopted after the 0.25 result was reported; 0.25 = the software default; the reported detector chose its checkpoint on the COBRA2026 and DeepFluoro test splits. [c]Non-interpolated average precision over the confidence sweep, computed for the design-B arms and the controls only. [d]Controls trained on the target device's own frames (test cases held out by case); on C-ArmLSpine the outline convention penalises these body-trained arms, which read 89.9 % (258/287) and 90.6 % (260/287) at 0.25 under body boxes. [e]One specimen is one cluster, no interval; the six-specimen rows test arms that never saw the device, their six-cluster intervals descriptive. [f]Scored at 0.05 and 0.25 only, on every annotated NHANES II lumbar spine that neither detector trained or was selected on; paired difference at 0.05 +31.7 percentage points [21.0, 45.4], sign test 16 vs 0 spines, Holm-adjusted $P < .001$; 17 clusters, so the intervals are approximate. [g]Wilson 95 % interval where the percentile one reached 0 or 100 % against the data (unclustered for the two 0 cells) or the file flags errors in few clusters (sacrum controls unadjusted, design effect about 1); Cios Fusion only: split by view, no pooled interval.

Table S3: Corpus by source: persons and primary images of the built person-disjoint split, transitional persons, and age and sex where a release gives them.

Source	Persons (train / val / test)	Images: real / rendered	Transitional persons[g]	Age, median [IQR] (n)	Female, % (n)	Age and sex read from
CTSpinoPelvic1K[b]	802 (636 / 73 / 93)	– / 802	52 (22); 64	59 [52–63] (709)	53.3 (738)	release manifest[d]
TotalSegmentator v2 and v3	1,867 (1,469 / 193 / 205)	– / 2,905	4 (3); 52	61 [44–73] (1,853)	42.5 (1,852)	meta.csv (v2.0.1, v3.0.0)
VerSe 19/20	355 (252 / 35 / 68)	– / 761	135 (52); 166	60 [49–73] (337)	51.0 (337)	organisers' demographics table
CSXA	4,963 (3,979 / 485 / 499)	4,963 / –	none	36 [30–46] (4,963)	64.5 (4,963)	descriptor table "Basic information"
BUU-LSPINE	3,598 (2,926 / 331 / 341)	7,203 / –	none	57 [41–67] (3,598)	63.3 (3,598)	dataset report, sheets Pos_LA / Pos_AP
NHANES II[c]	9,581 (7,673 / 969 / 939)	16,272 / –	none	55 [37–65] (9,581)	51.8 (9,581)	CDC public-use tape 5301, by sequence number
VinDr-SpineXR	4,981 (4,019 / 475 / 487)	10,274 / –	none	49 [38–59] (1,746)	62.3 (4,913)	DICOM headers (PatientAge, PatientSex)[d]
FluoroCap[e]	1,953 (excluded 1,953)	2,385 panels (excluded) / –	none	–	–	not read[e]
C-ArmLSpine[b]	149 (123 / 13 / 13)	299 + 301 sweep / –	3 (3); 4	none	none	none released[a]
DeepFluoro	6 (4 / 1 / 1)	366 / –	none	none	none	none released[a]
COBRA2026, Elekta XVI[b,c]	692 (445 / 71 / 159; hold 17)	3,480 / –	0 (0); 9	68 [58–75] (347)	43.9 (319)	metadata.yaml, ct block[d]
Geneva hip	1 phantom (19 views: 0 / 0 / 19)	116 / –	none	none	none	none released[a]
Ljubljana 2D-3D-GS	1 specimen, not in the split[b]	18 (no split root)[b] / –	none	none	none	none released[a]
AASCE[f]	217 (178 / 14 / 25)	440 / –	none	–	–	not read[f]
Spinal-AI2024[f,h]	44 (44 / 0 / 0)	– / 8,358	none	–	–	not read[f]
lateral236[f]	236 (193 / 22 / 21)	236 / –	none	–	–	not read[f]
Mendeley-716[e]	51 (excluded 51)	584 (excluded) / –	none	–	–	not read[e]

Persons are the built split's person keys (TotalSegmentator scans folded by content hash, no patient identifier; VinDr-SpineXR studies; BUU-LSPINE cases, two byte-identical pairs merged; NHANES II sample persons; COBRA2026 and C-ArmLSpine cases; Geneva, camera views of one phantom); hold = COBRA2026 cases held from every side. Images are the primary images of each source in the split's item map; other render roots (poses, styles, field-of-view cuts, film copies) re-render the same subjects and are not added. IQR = interquartile range. [a]No age or sex released: C-ArmLSpine birth dates are a 2020-01-01 placeholder with no age tag; DeepFluoro age and sex tags are placeholders (022Y on the sawbones too); Geneva is one phantom; the Ljubljana release files carry none. [b]Audit repairs: CTSpinoPelvic1K record 0533, a truncated download rebuilt after the banks were frozen, is a person and a render here (802) and in no trained bank (801, Table 1); C-ArmLSpine's 299 spot exposures are the count after the case134 repair (Table 1 scores 296), its 301 sweep projections are the one rotational acquisition, and its second released sweep and the 18 Ljubljana frames are in no split root; COBRA2026 frames are counted once after the duplicate-root repair. [c]Subsets Table 1 counts: the 364 annotated NHANES II persons read 65 [62–69] (364) and 55.2 % female (364); the 145 COBRA2026 detector cases 66 [56–73] (115) and 54.7 % (75). [d]"Other" and "unknown" sex codes are excluded from the female denominator (CTSpinoPelvic1K 11, VinDr-SpineXR 68 studies, COBRA2026 120 of 692); CTSpinoPelvic1K ages of 89 are the TCIA top-code (31 records); NHANES II releases no demographics with its films, and the sequence number in each file name joins the survey's public-use anthropometry tape, whose sex code was checked against median standing height; VinDr-SpineXR releases no table, both tags were read from every image header and no study's images disagreed. [e]Excluded: keyed by the split but in no bank and in no reported number (FluoroCap panels are figures of published articles; 500 of Mendeley-716's 584 films re-encode BUU-LSPINE films). [f]No Table 1 row; the release was not read for age or sex. [g]Transitional persons in the subtype table of 23 September 2026 that Table 2 was drawn from (test side in parentheses); after the semicolon, the transitional registry of 2026-09-24 over all truth tiers (read, two-labeller, single-labeller, morphology-only), whose labeller-tier calls are not certified persons and are not in Table 2. [h]Machine-generated films. Source files: results/age_sex_by_source.json (generated 2026-09-24 20:34:35, SHA-256 1882037348892521) and results/corpus_census.json (computed 2026-09-24 20:32 from the split of 2026-09-23 20:18:48, SHA-256 65a57eb0c7a6e202; file SHA-256 ded51200361f4e23).

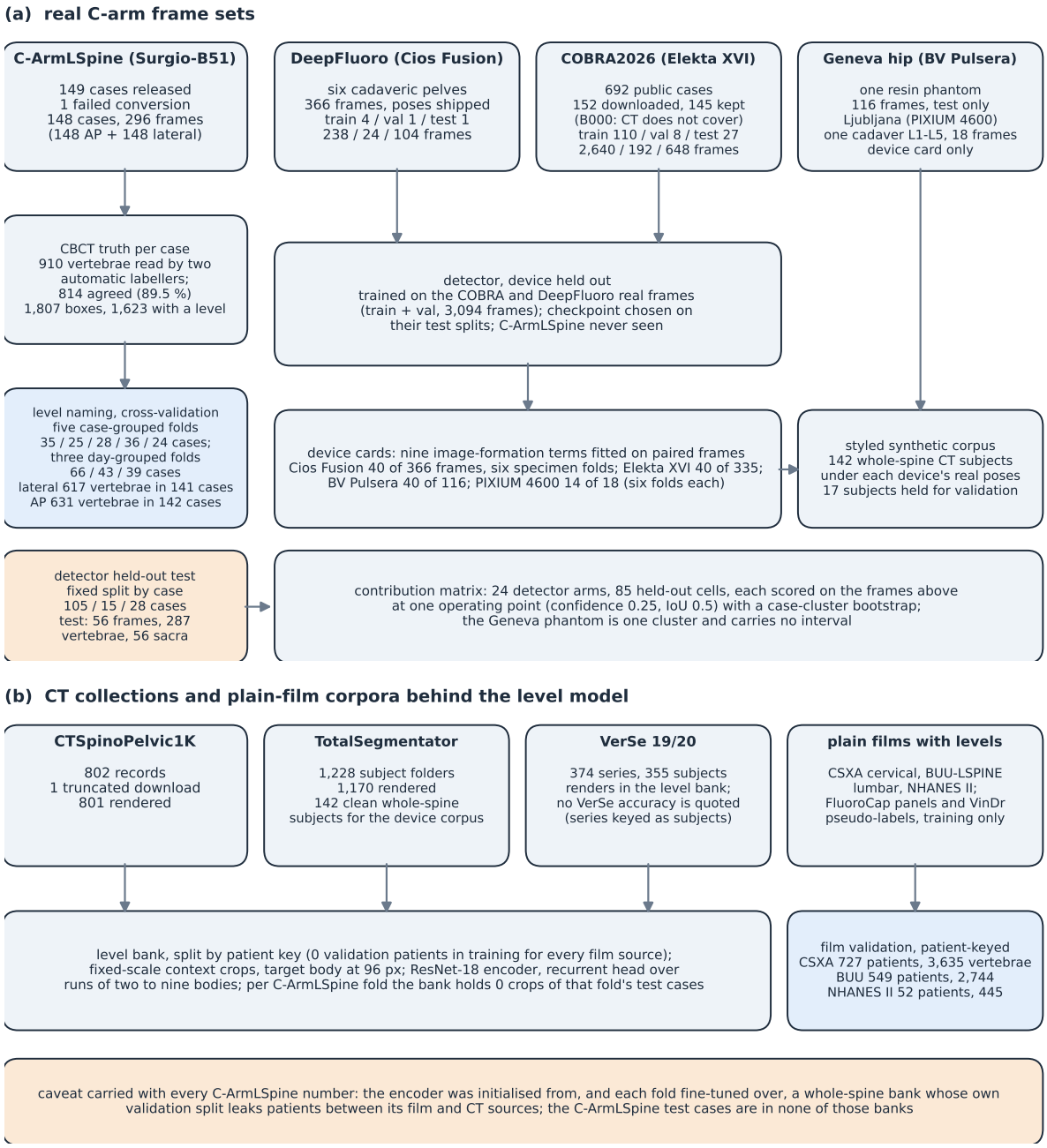
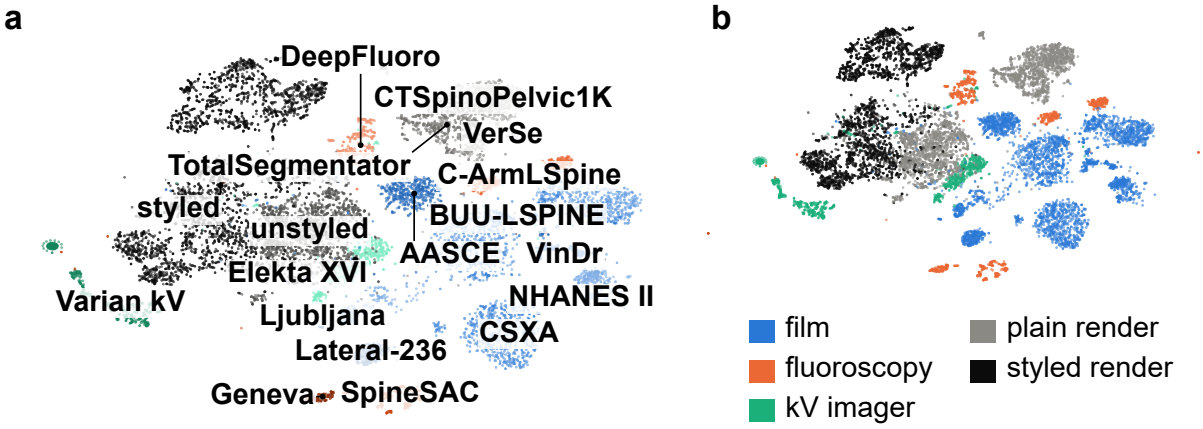
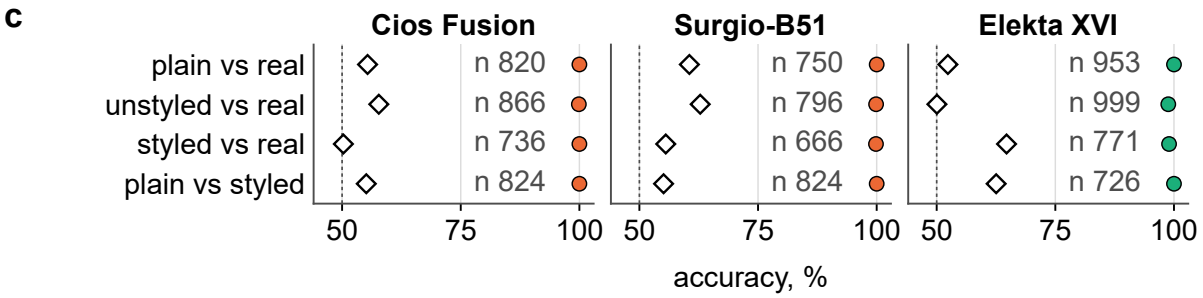


Figure S1: Flow of cases. (a) Real C-arm frame sets, their splits and what each was used for. (b) CT collections and plain-film corpora behind the level model, with the patient-keyed film validation split and the caveat carried with every C-ArmLSpine number.



Film vs C-arm and kV imager, one source held out, n = 5,797: majority share 64.4 %; image statistics 63.4 % [57.6, 69.2], deep features 85.0 % [79.4, 90.0]



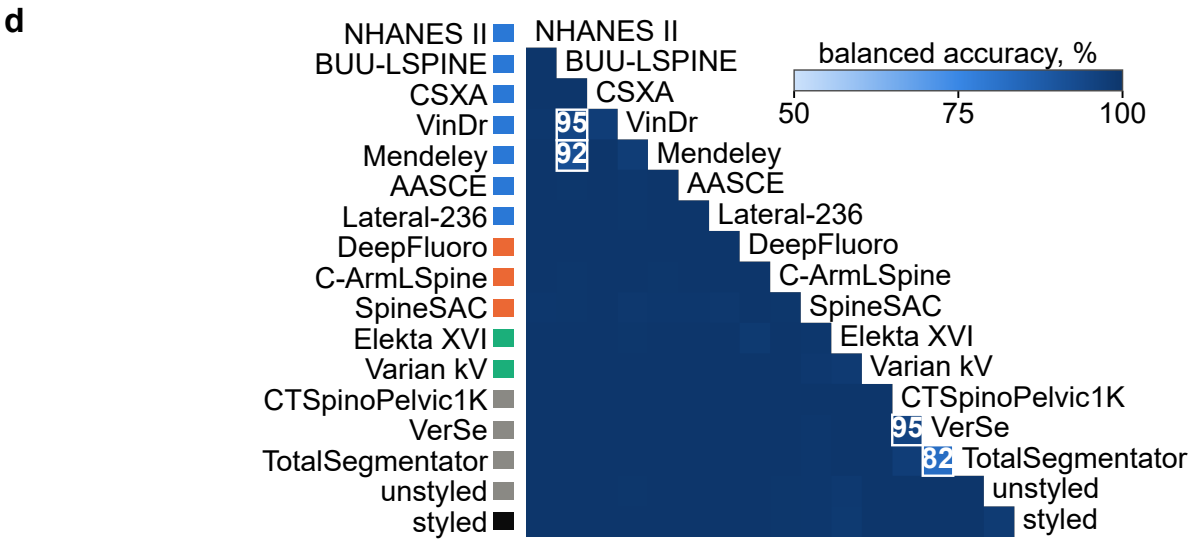


Figure S2: Corpus manifold, ImageNet features. (a) t-SNE (t-distributed stochastic neighbour embedding) of 11,638 frames from 18 sources, one colour family per modality. (b) The same points by modality; below, a film-versus-C-arm classifier. (c) Each render family against its device's real frames: accuracy, 95 % case-cluster interval; diamonds, majority share. (d) Pairwise balanced accuracy; Mendeley only here.

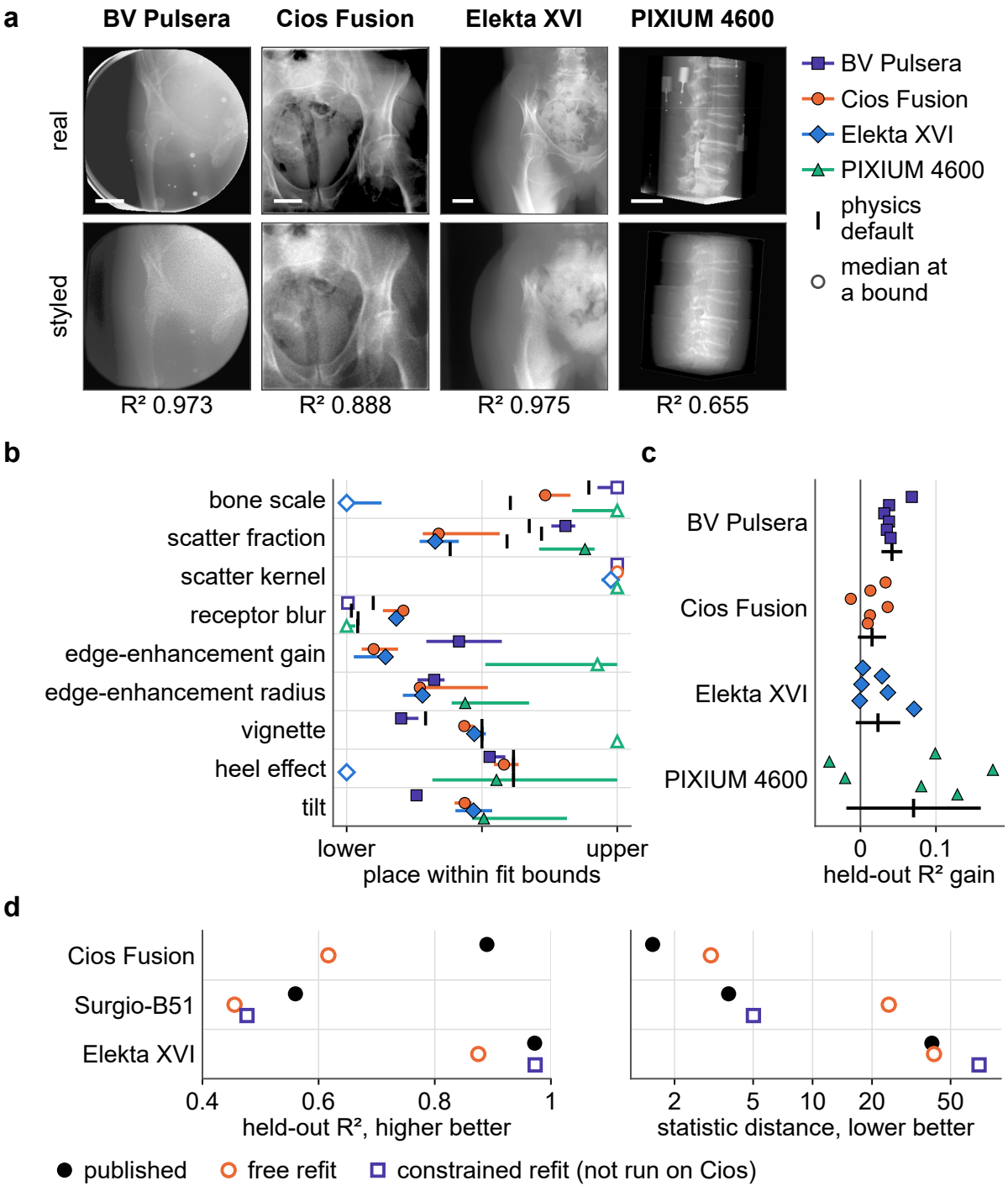


Figure S3: Image-formation parameters. (a) Real frame per device above its styled render; bar, 50 mm; R^2, body-region fit. (b) Fitted terms within their bounds: fold median, range. (c) Per-fold R^2 gain over uncalibrated physics (Pulsera, PIXIUM within one object); bar, six-fold t interval. (d) A refit with image noise in the loss failed its own gate; published parameters stand.

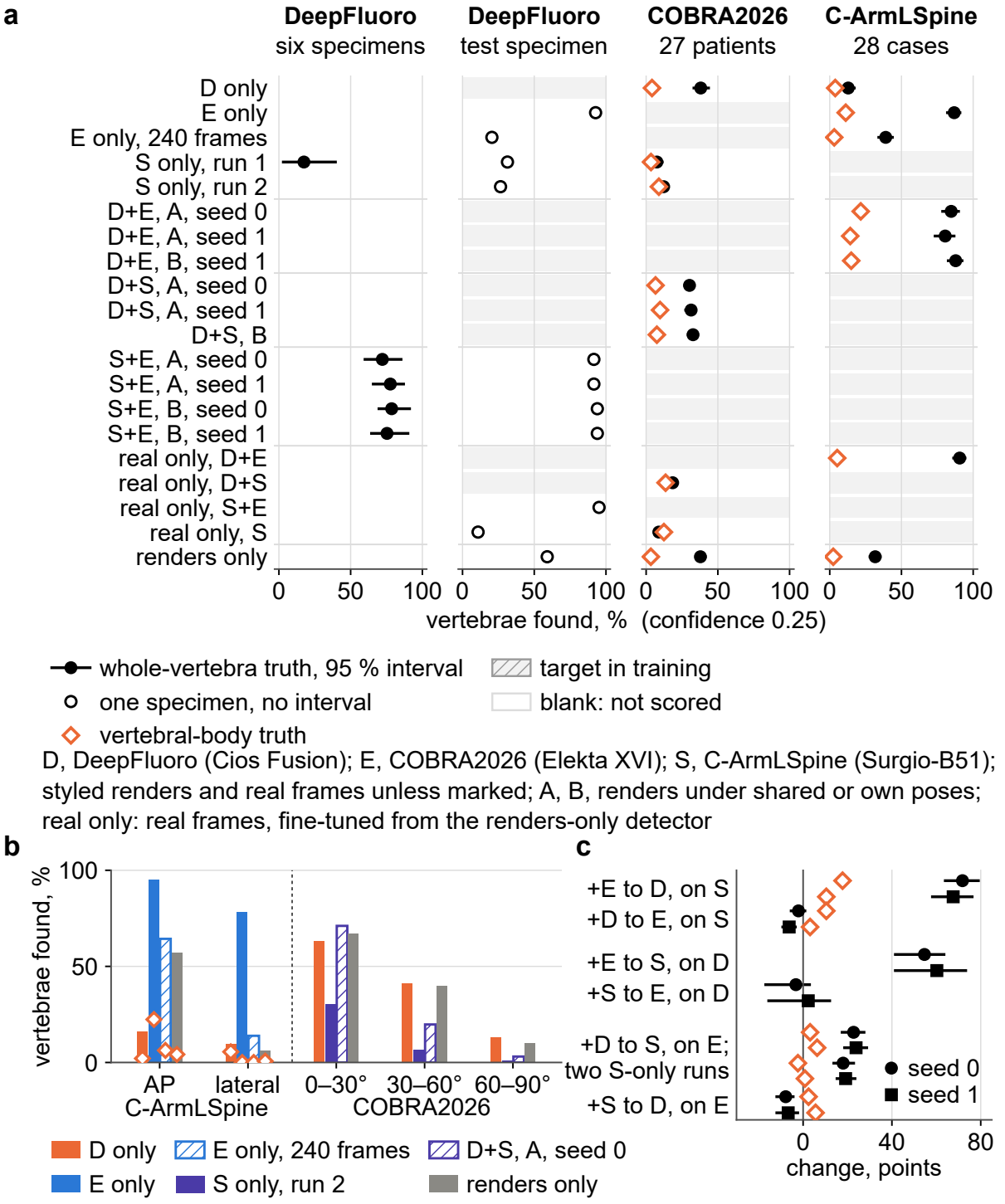


Figure S4: Which training source carries the detector. (a) Held-out vertebra recall per training arm and target: dots, whole-vertebra truth, 95 % case-bootstrap interval; diamonds, the same predictions against vertebral-body truth, a box-convention reading, not a retrained score. (b) Recall by view or gantry-angle band. (c) Recall change from adding one source, tested on the third; diamonds, body-truth difference, no interval.

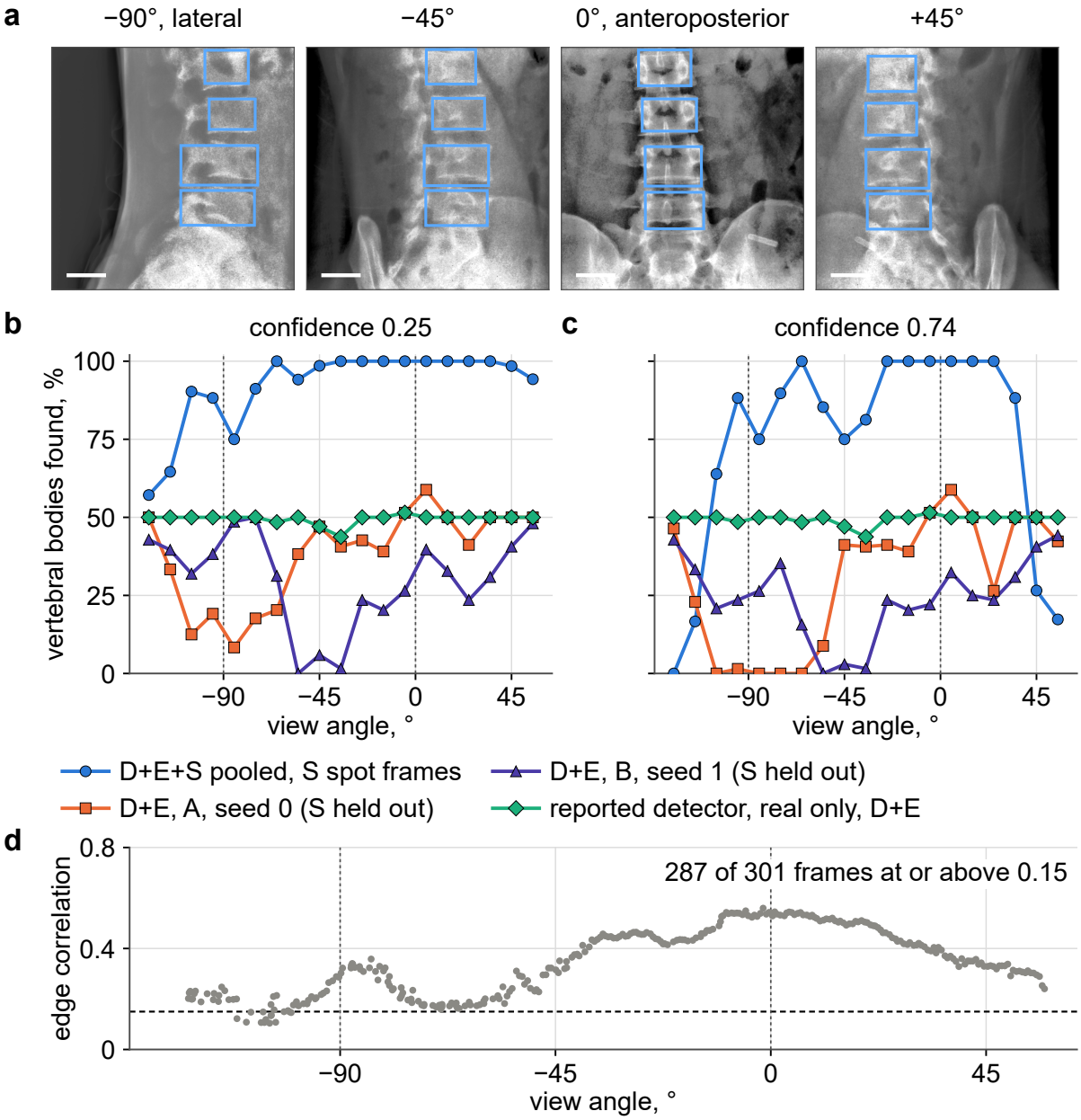


Figure S5: Detection against view angle on one patient's real rotational C-arm acquisition (301 frames; no intervals). (a) Lateral (−90°, source at patient's right) to +45°; boxes, bodies projected from cone-beam CT; bar, 50 mm. (b, c) Bodies found per 10° bin; arm names as in Figure S4; the reported detector's flat 50 % is unexplained. (d) Registration quality.